

Toric methods for Calabi–Yau compactifications and the string landscape

Lecture notes for ESI 2026 programme on

*The Unreasonable Effectiveness of Toric Geometry:
Bridging Mathematics, Computation, and String Theory*

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Scope and lecture outline

These lectures are an introduction to **toric methods for Calabi–Yau compactifications and the string landscape** for mathematicians. The organising question is concrete: how much of a four-dimensional effective theory, and in particular of its vacuum energy, can be computed from explicit algebraic and toric data? The candidate de Sitter construction of [1] is the central worked example, but the first goal is more basic: to explain why explicit Calabi–Yau geometry is needed at all.

The basic toric vocabulary (fans, cones, divisors, the secondary fan etc.) is used throughout without being re-derived in the live lectures; it is collected in the appendices. By contrast, the constructions that are central to the worked example, in particular Batyrev’s construction of Calabi–Yau hypersurfaces from reflexive polytopes and the role of FRSTs, are re-derived at the level of a stated theorem in Lecture 1 (see Theorem 1.12). The appendices collect the definitions and standard results from algebraic geometry, convex geometry, toric geometry, mirror symmetry, orientifolds, and computational workflows that are needed to read the notes as a self-contained reference.

Sections marked by an asterisk are included for completeness and for later reference, but they are not part of the intended blackboard delivery. In the live lectures I will use them only as optional background or as answers to questions.

Prerequisites.

- Algebraic and complex geometry at the level of Huybrechts (*Complex Geometry* [2]) or Voisin (*Hodge Theory and Complex Algebraic Geometry* [3]).
- Toric geometry at the level of Cox–Little–Schenck [4]: fans, divisors, intersection rings, the Kähler and Mori cones, the secondary fan, and reflexive polytopes. Batyrev’s construction [5] and the Kreuzer–Skarke list [6] are helpful background but are recalled in Lecture 1 (see Section 1.9).
- Riemannian and Hermitian geometry: holonomy and the Hodge decomposition. Berger’s classification is helpful background but is restated in Lecture 1 (see Theorem 1.2).
- Basic physics vocabulary is helpful but not assumed: fields, actions, equations of motion, compactification, fluxes, branes, and effective field theory are introduced when first used.

Motivation. The motivation is two-fold. First, the notes explain how toric geometry makes certain string compactifications explicit enough that one can compute physical quantities rather than only discuss them abstractly. The conceptual ingredients have long been established: Calabi–Yau geometry, mirror symmetry, Picard–Fuchs equations, Gopakumar–Vafa invariants, D-branes, and four-dimensional supergravity have all been studied for decades. What has changed is the feasible computational range. Many calculations that were once limited to one- or few-modulus examples can now be attempted for the largest known Hodge numbers, i.e., compactifications with hundreds of moduli, provided one keeps careful control of truncations and chamber structure.

Second, the notes are meant to identify bottlenecks. The concrete questions are geometric before they are physical. Given a Calabi–Yau threefold X , how does one determine its Kähler cone, effective cone etc.? How does one compute minimal-volume representatives of divisor classes, especially when holomorphic representatives exist only after moving in the space of birational equivalence classes of X ? Which effective divisors have $h^\bullet(D, \mathcal{O}_D) = (1, 0, 0)$ and can therefore contribute to the non-perturbative superpotential? How should one compare the Wall data of hypersurfaces arising from different triangulations, and how many inequivalent geometries are really present in the Kreuzer–Skarke ensemble? The hope is that the lectures make these questions precise enough to be useful to both the mathematics and string theory communities.

Lecture outline. The live blackboard narrative is deliberately narrower than the written notes: compactification first selects Ricci-flat internal geometries; the deformations of those geometries become four-dimensional moduli; fluxes, sources and quantum corrections generate a scalar

potential V ; and toric geometry supplies much of the discrete data needed to write and analyse that potential. The detailed explanation of how ten-dimensional fields reduce to four-dimensional fields appears only after the Calabi–Yau moduli have been introduced. In this sense the purpose is not to give a general survey of string theory, but to explain why explicit geometry is needed to compute its four-dimensional consequences.

There is intentional overlap with the TASI lectures [7]. The TASI notes are the physics-first reference for the Type IIB flux compactification, the leading-order effective field theory, and the candidate de Sitter construction. The present notes reorganise that material for the ESI audience: the blackboard route starts from compact Ricci-flat geometry, then explains which toric computations supply the input to the four-dimensional scalar potential. Whenever a statement is imported from the TASI treatment rather than rederived here, the relevant TASI section is cited explicitly.

1. **Lecture 1: The why and how of Calabi–Yau.** The first lecture introduces the minimal string theory input, explains why compactification leads to a Ricci-flat internal geometry, why Calabi–Yau threefolds are the computable and best understood class used in these notes, and how Batyrev’s construction turns reflexive polytopes and FRSTs into explicit Calabi–Yau hypersurfaces in toric varieties. The emphasis is pragmatic: what toric methods compute reliably today (Hodge data, intersection numbers, cones, divisor topology), and what remains beyond current algorithms. Along the way we preview where these data enter physics: flux superpotentials, instanton corrections, divisor zero-mode tests, and Kähler-cone navigation.
2. **Lecture 2: Periods and GV invariants from toric data.** The second lecture introduces the minimal Type IIB/O3–O7 dictionary needed to write the flux superpotential and then focuses on the main computational problem:

$$W_{\text{flux}} = \int_X \Omega \wedge G_3. \quad (0.1)$$

Fluxes supply the integral coefficients; periods supply the functions. The lecture explains how mirror symmetry, toric hypersurfaces, the mirror map, and genus-zero GV invariants make these periods computable. It then uses PFVs as the final application: special integral fluxes make the polynomial LCS contribution flat, while the GV-controlled instanton terms lift the remaining direction and produce a computable small value of W_0 . Non-perturbative Kähler-modulus stabilisation and the candidate-dS pipeline are deferred to Lecture 3.

3. **Lecture 3: Vacuum energies from toric geometry.** The third lecture assembles the leading-order effective theory for Type IIB O3/O7 orientifold compactifications with flux, non-perturbative Kähler-modulus stabilisation, and a redshifted antibrane uplift. It introduces geometric phases, flop transitions, vex chambers, and the extended Kähler cone, then explains how chamber-aware algorithms, including `fanroots` and `JAXVacua`, locate Kähler moduli critical points. The remaining ingredients are the Klebanov–Strassler throat, KPV anti-D3 uplift, the explicit toric AdS precursor and candidate-dS recipes, and the control checks: tadpoles, GV convergence, higher-order corrections, backreaction, and quantisation issues. The list is retained because these checks distinguish the leading-order candidate construction from a fully controlled vacuum theorem.

1 Lecture 1: The why and how of Calabi–Yau

Anchor: TASI [7], §§ 1–2.

Guiding question. What can toric methods compute for string compactifications, and where do current methods still fall short?

The answer will be deliberately two-sided. Toric geometry gives an explicit, computable supply of Calabi–Yau threefold hypersurfaces: Hodge numbers, intersection forms, cones, divisor topology, periods, and enumerative inputs such as Gopakumar–Vafa invariants can be extracted algorithmically. It does not, by itself, solve the Ricci-flat metric, determine every effective curve or divisor on the hypersurface, or provide access to all quantum corrections.

Historical and computational thread. The geometric problem starts with two classical inputs. Yau’s solution of the Calabi conjecture implies the existence of Ricci-flat Kähler metrics on compact Kähler manifolds with trivial canonical bundle [8], while Calabi–Yau threefolds entered string compactification through the work of Candelas, Horowitz, Strominger and Witten [9]. The reason these lectures focus on toric hypersurfaces is more computational: Batyrev’s reflexive polytopes give a large, explicit class of mirror pairs [5]; the Kreuzer–Skarke classification makes this class finite and searchable [6]; and modern packages such as `CYTools` turn the resulting combinatorial data into the intersection, cone and divisor-topology inputs used later [10]. Lecture 1 explains enough of this chain to make Lectures 2 and 3 self-contained.

Goals. After this lecture you can

1. state what a string compactification is, as a geometric object, and why the vacuum Einstein equations on a product ansatz select Ricci-flat internal manifolds;
2. explain why Calabi–Yau threefolds are the computable Ricci-flat class used in these lectures, and how reduced holonomy gives the parallel spinors responsible for residual supersymmetry;
3. identify the moduli of a Calabi–Yau compactification with specific cohomological data;
4. explain why Calabi–Yau hypersurfaces in toric varieties are the natural constructive source of explicit compact examples, and what data they feed into Lectures 2 and 3.

1.1 Why string theory, and why compactify?

Let us motivate how geometry enters physics. We have very successful but structurally different theories for matter and gravity:

$$\begin{array}{ll}
 \text{matter} & : \quad \text{Standard Model,} \\
 & \quad \text{language: quantum field theory,} \\
 & \quad \text{hypothesis: point particles;} \\
 \text{gravity} & : \quad \text{General Relativity,} \\
 & \quad \text{language: differential geometry,} \\
 & \quad \text{hypothesis: spacetime as a manifold.}
 \end{array} \tag{1.1}$$

The quantum gravity problem is to find a framework in which these two descriptions can be combined. Perturbative quantum gravity based on point particles is not perturbatively renormalisable, and curvature singularities indicate that the classical spacetime description itself can break down. String theory is one proposed framework in which the fundamental extended objects are one-dimensional strings rather than point particles.

The useful contrast is the following. A particle traces out a worldline in spacetime; a string traces out a worldsheet Σ , and the simplest classical action is proportional to its area,

$$S_{\text{NG}} = -T \int_{\Sigma} dA, \tag{1.2}$$

where T is the string tension. Quantising the string gives a tower of vibrational modes. In the point-particle picture one quantises the possible motions of a point; in the string picture one also quantises the normal modes of an extended object. Very schematically,

$$\text{particle: worldline} \quad \longrightarrow \quad \text{string: worldsheet with oscillator modes.} \tag{1.3}$$

The word “harmonics” here refers to these oscillator modes of the string. After quantisation they appear, from the spacetime point of view and for an observer sufficiently far away, as different particle species: scalars, gauge fields, spinors, and a spin-2 graviton among them.

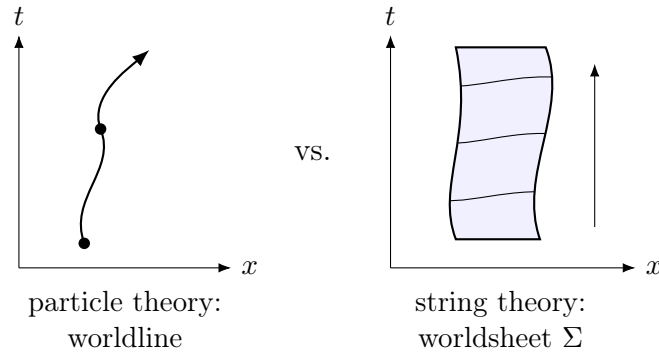


Figure 1: A point particle sweeps out a one-dimensional worldline, whereas a string sweeps out a two-dimensional worldsheet. In both panels the object evolves upward in target time t : the point traces a curve, while the spatially extended string traces a strip.

It is useful to separate two attitudes. One can use string theory and related constructions as mathematical tools in broader effective descriptions. In these lectures we use the stronger quantum gravity interpretation: the theory is required to include gravity, and the perturbative spectrum contains a spin-2 graviton. With this interpretation, the consistency conditions of the quantum theory are part of the input. In particular, for the critical superstring the consistency of the quantum theory fixes the target spacetime dimension to ten. At energies small compared with the string scale, one keeps only the massless modes. The massive oscillator modes have masses of order the string scale M_s , so a low-energy observer with characteristic energy $E \ll M_s$ cannot excite them as dynamical fields. Their effects are still present indirectly, through higher-derivative corrections to the effective action. The one fact from this construction that Lecture 1 needs is that the massless spectrum of a consistent superstring includes a ten-dimensional metric, together with further geometric fields such as differential forms, scalars, spinors, and gauge fields. The metric is what will make Ricci-flat geometry enter immediately.

Since observed physics is effectively four-dimensional at accessible energies, a compactification must explain how a ten-dimensional theory can give a four-dimensional effective theory:

$$\dim \mathcal{M}_{10} = 10 = 9 + 1, \quad \dim \mathcal{M}_4 = 4 = 3 + 1, \quad \mathcal{M}_{10} \simeq \mathcal{M}_4 \times X. \quad (1.4)$$

The compact six-dimensional factor X is where geometry enters. The central point of the lectures is that the topology and geometry of X determine the light four-dimensional fields, their couplings, and the scalar potential whose critical values are the vacuum energies.

1.2 Framing: what a string compactification is

For the first lecture, the input from string theory can be stated without choosing a specific string theory. It is the following geometric package.

1. A *ten-dimensional spacetime* $\mathcal{M}_{10}$: a Lorentzian 10-manifold with metric g_{MN} .
2. Additional *fields* Φ on $\mathcal{M}_{10}$. Here a field simply means a geometric object varying over spacetime: a section of a vector bundle, such as a differential form, a function, a connection, a spinor, or the metric itself. In Lecture 1 we keep these fields mostly silent and set them to zero or to constants when deriving the Ricci-flatness condition.
3. *Source terms and charged objects*, such as branes and orientifold planes, which enter through $\mathcal{L}_{\text{source}}$ and produce localised stress-energy and charge. A brane is an extended object on which open strings can end, or more generally an object carrying charge under one of the differential-form gauge fields of the theory. Its worldvolume is a submanifold of spacetime, so after compactification a brane can wrap a cycle in X and fill some or all of the visible

four-dimensional spacetime. This is why branes are simultaneously geometric objects and physical sources: they contribute localised energy-momentum, carry quantised charge, and can support gauge fields or matter in the four-dimensional theory [11].

4. An action functional

$$S[g, \Phi] = \frac{1}{2\kappa_{10}^2} \int_{\mathcal{M}_{10}} d^{10}x \sqrt{-g} \left(R^{(10)} + \mathcal{L}_{\text{matter}}(g, \Phi) + \mathcal{L}_{\text{source}}(g, \Phi) + \underbrace{\quad}_{\text{e.g. } R^4} \right), \quad (1.5)$$

whose Euler–Lagrange equations are the equations of motion, i.e., the differential equations obtained by requiring the first variation of S to vanish, see Eq. (1.9) below. This is the classical, leading-order ten-dimensional action. The full string effective action also contains higher-derivative and quantum corrections.¹ We do not include those corrections in the first-pass vacuum equation, but they reappear in Lecture 2 as corrections to the four-dimensional effective action.

Let x^μ be coordinates on a four-dimensional Lorentzian manifold $\mathcal{M}_4$ and y^m coordinates on a compact real 6-manifold X . A warped product metric on $\mathcal{M}_{10} = \mathcal{M}_4 \times X$ has the form

$$ds_{10}^2 = e^{2A(y)} g_{\mu\nu}^{(4)}(x) dx^\mu dx^\nu + e^{-2A(y)} g_{mn}^{(6)}(y) dy^m dy^n. \quad (1.6)$$

The function $A(y)$ is the *warp factor*. In the first pass through the vacuum equations we set $A = 0$, so the metric is the ordinary product metric $ds_{10}^2 = ds^2(\mathcal{M}_4) + ds^2(X)$. This is the Kaluza–Klein bridge in its simplest form: a four-dimensional observer does not see the internal coordinates directly, but sees the coefficients of expansions in functions, forms, and metric deformations on X .

Definition 1.1 (String compactification, geometrically). *A string compactification consists of:* (i) *a factorisation $\mathcal{M}_{10} = \mathcal{M}_4 \times X$ of the ten-dimensional spacetime into a four-dimensional Lorentzian manifold $\mathcal{M}_4$ and a compact 6-dimensional Riemannian manifold X (the internal space);* (ii) *a solution of the ten-dimensional equations of motion on $\mathcal{M}_{10}$ with metric in the warped product form (1.6);* (iii) *background choices for any non-metric fields and charged objects that are present, subject to the global consistency conditions of the chosen string theory.*

On a compact internal space, closed differential-form field strengths can have quantised periods. Abstractly, if F_p is a closed p -form field strength, *flux quantisation* means that its normalised periods over integral p -cycles are integers, so the background defines an integral cohomology class. The precise normalisation and even the correct cohomology theory depend on the string theory and on the charged objects present. Lecture 2 specialises this statement to Type IIB, where the flux superpotential uses the three-form classes $[F_3], [H_3] \in H^3(X, \mathbb{Z})$.

At this point the lecture has only defined the geometric problem: choose a compact six-manifold X , a metric on $\mathcal{M}_4 \times X$, and background fields and sources so that the ten-dimensional equations of motion are satisfied, and then determine the four-dimensional fields and interactions obtained by expanding around that solution. We now first find the internal geometries that can solve the equations of motion in the compactification background, and then ask which four-dimensional fields and potentials those geometries produce.

1.3 The vacuum solution problem

We begin in the simplest setting: no fluxes, no branes. We ask which Riemannian metrics on X are compatible with a vacuum solution.

¹The first well-known perturbative correction to the ten-dimensional type-II action contains an R^4 term at order $(\alpha')^3$. The parameter α' is the inverse string tension, up to the conventional factor 2π , and sets the square of the string length. Corrections in powers of α' are higher-derivative or finite-string-size corrections. The string coupling g_s controls the loop expansion.

The metric equation comes from varying the schematic action above with respect to g^{MN} . Let

$$S_{\text{matter+source}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g} (\mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{source}}), \quad (1.7)$$

where $\mathcal{L}_{\text{matter}}$ contains the kinetic terms of the non-metric fields Φ and $\mathcal{L}_{\text{source}}$ contains localised brane/orientifold terms. Define

$$T_{MN} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter+source}}}{\delta g^{MN}}. \quad (1.8)$$

Then the ten-dimensional Einstein equation is

$$R_{MN}^{(10)} = \kappa_{10}^2 \left(T_{MN} - \frac{1}{8} g_{MN} T^K{}_K \right). \quad (1.9)$$

In this first pass we set all additional fields to zero or constant and include no localised sources, so $T_{MN} = 0$. The unwarped product version of (1.6) then gives

$$R_{\mu\nu}^{(4)} = 0, \quad R_{mn}^{(6)} = 0. \quad (1.10)$$

The four-dimensional Einstein vacuum equation forces $\mathcal{M}_4$ to be Ricci-flat. For our purposes the physically interesting case is $\mathcal{M}_4 = \mathbb{R}^{1,3}$ (Minkowski), so $R_{\mu\nu}^{(4)} = 0$ holds trivially. The interesting condition is the internal one: *X must be Ricci-flat*.

The problem is then: construct compact Ricci-flat Riemannian 6-manifolds on which one can compute. This is a hard nonlinear PDE problem: explicit Ricci-flat metrics on compact non-flat 6-manifolds are generally not known in closed form. The useful insight is that one does not need the metric explicitly in order to build the compactification. Yau’s theorem guarantees the *existence* of Ricci-flat Kähler metrics from complex-geometric data, and toric geometry gives a large explicit class of spaces for which the relevant topological data are computable.

1.4 Berger’s holonomy classification

The compactification problem above asks for Ricci-flat internal manifolds with enough structure to control the low-energy theory. Holonomy is the efficient invariant for this purpose. It records how tangent vectors, tensors and spinors transform under parallel transport, and therefore tells us which parallel geometric objects a metric can admit.

Theorem 1.2 (Berger, holonomy classification [12, 13]). *Let (X, g) be a simply-connected, complete Riemannian manifold whose holonomy representation is irreducible and whose metric is not locally symmetric. Then the (restricted) holonomy group $\text{Hol}(X, g) \subseteq \text{SO}(\dim X)$ is one of groups in Table 1. The lower bounds $m \geq 2$ remove degenerate or duplicated low-dimensional cases ($\text{U}(1) = \text{SO}(2)$, $\text{Sp}(1) = \text{SU}(2)$, $\text{Sp}(1) \cdot \text{Sp}(1) = \text{SO}(4)$).*

Berger’s theorem is useful here because it sharply limits the possible irreducible, non-symmetric Riemannian holonomies. It does *not* classify all compact Ricci-flat manifolds. Rather, it explains why, once one asks for a six-dimensional irreducible Ricci-flat background admitting parallel spinors, the natural holonomy group is contained in $\text{SU}(3)$. In the terminology of [7], the compact Ricci-flat examples used in string compactification all arise from reduced, or special, holonomy; the Calabi–Yau case is the six-dimensional member of this list that is complex, Kähler, and computationally accessible. This is a guide to the known constructive landscape, not a theorem excluding every possible Ricci-flat six-manifold outside the classes we discuss.

For a compact Kähler threefold, the algebraic-geometric condition that implements the $\text{SU}(3)$ holonomy target is the triviality of the canonical bundle. Let $\mathcal{O}_X$ denote the structure sheaf of holomorphic functions on X ; an isomorphism $K_X \cong \mathcal{O}_X$ means that the canonical bundle is holomorphically trivial, equivalently that there is a nowhere-vanishing holomorphic $(3, 0)$ -form.

Holonomy	Dimension	Ricci-flat?	Geometry relevant here
$\mathrm{SO}(n)$	n	no generic implication	no parallel spinor in general
$\mathrm{U}(m)$	$2m$	not necessarily	Kähler
$\mathrm{SU}(m)$	$2m$	yes	Calabi–Yau (Kähler+Ricci-flat)
$\mathrm{Sp}(m)$	$4m$	yes	hyperkähler
$\mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$	$4m$	no (Einstein)	quaternionic Kähler
G_2	7	yes	G_2 manifolds
$\mathrm{Spin}(7)$	8	yes	$\mathrm{Spin}(7)$ manifolds

Table 1: Berger’s list of irreducible Riemannian holonomies.

Holomorphic triviality of K_X gives $c_1(X) = 0$; Yau’s theorem then gives a Ricci-flat Kähler metric in each Kähler class; and, after excluding flat or product factors, the holonomy is the expected subgroup of $\mathrm{SU}(3)$,

$$K_X \cong \mathcal{O}_X \implies \text{Ricci-flat Kähler metric} \implies \mathrm{Hol}(X) \subseteq \mathrm{SU}(3). \quad (1.11)$$

When the holonomy is exactly $\mathrm{SU}(3)$, this is the generic irreducible Calabi–Yau threefold case. Proper subgroups correspond to extra structure, such as flat factors or product geometries, and usually to more supersymmetry than the generic threefold case. This is the precise sense in which Calabi–Yau threefolds are the workhorse Ricci-flat backgrounds of the lectures: they are explicit enough to construct and compute with, and their reduced holonomy has the spinorial consequence needed by the physics.

1.5 Aside: Reduced holonomy, parallel spinors, and supersymmetry*

Supersymmetry is not the starting point of the geometric argument; it is a consequence of the reduced holonomy just described. The physics reason one cares about it is that residual supersymmetry gives technical control of the low-energy theory and organises the four-dimensional fields into multiplets. The mathematical statement is the existence of parallel sections of the spinor bundle.

Definition 1.3 (Parallel (covariantly constant) spinor). *Let (X, g) be a Riemannian spin manifold with spinor bundle $S \rightarrow X$ and let ∇ denote the spin connection. A parallel (or covariantly constant) spinor is a smooth section $\eta \in \Gamma(S)$ with $\nabla\eta = 0$.*

Remark 1.4 (Terminology). In the physics literature these are often called *Killing spinors*. In the standard differential-geometry convention, a *Killing spinor* satisfies $\nabla_X \eta = \lambda X \cdot \eta$ for a nonzero constant λ , and the $\lambda = 0$ case is called a *parallel spinor* and treated separately. For the unwarped Minkowski compactification considered here the relevant case is $\lambda = 0$, and we use the term *parallel* throughout.

Proposition 1.5 (Physics dictionary; standard supergravity Killing-spinor analysis, [7] § 2.3). *For the configuration $\mathcal{M}_4 \times X$ with $\mathcal{M}_4 = \mathbb{R}^{1,3}$, the unwarped product metric, no fluxes, and no branes, vanishing of the supergravity SUSY variations on the four-dimensional vacuum requires that X admit a parallel spinor. Starting from a ten-dimensional Type II theory with thirty-two real supercharges, one independent complex parallel spinor on X corresponds to $\mathcal{N} = 2$ four-dimensional supersymmetry (eight real supercharges) before any orientifold projection; starting instead from a heterotic or Type I theory (sixteen real supercharges in ten dimensions), the same condition yields $\mathcal{N} = 1$ in four dimensions (four real supercharges), as in Candelas–Horowitz–Strominger–Witten [9].*

Status: *this is a standard physics derivation, not a mathematical theorem about (X, g) alone, and assumes the leading-order supergravity expansion.*

The invariant internal spinors determine the residual supersymmetry. The precise conversion between internal spinor multiplicities and four-dimensional supercharges depends on the ten-dimensional chirality and real/complex spinor conventions. We only use the standard Type IIB case: an $SU(3)$ -holonomy Calabi–Yau threefold gives $\mathcal{N} = 2$ supersymmetry in four dimensions (eight real supercharges) before the orientifold projection in Lecture 2 reduces it to $\mathcal{N} = 1$.

The bridge to topology is Berger’s classification of holonomies on Riemannian manifolds. The curvature condition comes from the commutator of spin covariant derivatives. In an orthonormal frame,

$$\nabla_m \eta = \partial_m \eta + \frac{1}{4} \omega_m^{ab} \Gamma_{ab} \eta, \quad [\nabla_m, \nabla_n] \eta = \frac{1}{4} R_{mn}^{ab} \Gamma_{ab} \eta. \quad (1.12)$$

If η is parallel, the left-hand side vanishes, hence

$$R_{mn}^{ab} \Gamma_{ab} \eta = 0. \quad (1.13)$$

By the Ambrose–Singer theorem the restricted holonomy algebra is generated by curvature endomorphisms, so this integrability condition says precisely that the holonomy representation fixes a non-zero spinor. This can happen on a non-flat manifold only if the holonomy group acts trivially on some subspace of the spinor representation; see [14] for the spin geometry and [7], § 2.3 for the physics dictionary.

1.6 Calabi–Yau threefolds: existence and Hodge structure

The six-dimensional geometry problem has now become concrete: find compact Riemannian 6-manifolds whose holonomy is contained in $SU(3)$, and preferably equals $SU(3)$ after excluding torus and product factors. The systematic way to produce such spaces is to construct compact Kähler threefolds with trivial canonical bundle and then apply Yau’s theorem to obtain Ricci-flat metrics. Toric Calabi–Yau hypersurfaces are not meant to exhaust the known compact examples with $SU(3)$ holonomy. Rather, Calabi–Yau threefolds are the standard geometric class relevant to this compactification problem, and Batyrev hypersurfaces are the large computable family used in these lectures.

Definition 1.6 (Calabi–Yau threefold). *A Calabi–Yau threefold is a compact Kähler manifold X of complex dimension 3 with trivial canonical bundle $K_X \cong \mathcal{O}_X$ and $\pi_1(X) = 0$.*

Equivalently, X admits a nowhere-vanishing holomorphic $(3, 0)$ -form Ω , unique up to scale. The appendix records the standard comparison with other conventions, including formulations in terms of $c_1(X)$ and holonomy.

The simple-connectedness clause excludes T^6 , $K3 \times T^2$, abelian threefolds, and CY3s obtained as free quotients by a finite group with nontrivial π_1 (e.g. the heterotic Standard-Model quotients of Donagi–Ovrut [15] and Bouchard–Donagi [16]). It implies $h^{1,0}(X) = 0$, removing spurious vector multiplets from $H^{1,0}$ and giving a clean four-dimensional $\mathcal{N} = 1$ truncation in Lecture 2. The converse fails in general: free finite quotients can have torsion π_1 while still satisfying $h^{1,0} = 0$. The rigorous classification of compact Kähler manifolds with $c_1(X) = 0$ in $H^2(X, \mathbb{R})$ is given by the Beauville–Bogomolov decomposition [17, 12]: such an X admits a finite étale cover that is a product of complex tori, strict Calabi–Yau factors (with $\text{Hol} = SU(n)$, $n \geq 3$), and irreducible hyperkähler factors (with $\text{Hol} = Sp(n)$).

Theorem 1.7 (Yau [8]). *Let X be a compact Kähler manifold with $c_1(X) = 0$. Then in each Kähler class $[J] \in H^{1,1}(X) \cap H^2(X, \mathbb{R})$ there exists a unique Kähler metric g with $\text{Ric}(g) = 0$.*

Corollary 1.8. *Calabi–Yau threefolds admit Ricci-flat Kähler metrics; in particular they solve the vacuum Einstein equation $R_{mn}^{(6)} = 0$ required by the unwarped product ansatz.*

What follows is the cohomological data we will need in Lecture 2:

Proposition 1.9 (CY3 Hodge diamond). *Let X be a Calabi–Yau threefold. The only independent Hodge numbers are $h^{1,1}$ and $h^{2,1}$, and the Euler characteristic is $\chi(X) = 2(h^{1,1} - h^{2,1})$.*

The holomorphic $(3,0)$ -form Ω is unique up to overall rescaling and is nowhere vanishing. The pair $(h^{1,1}, h^{2,1})$ is the principal topological invariant of X for our purposes; we will see that $h^{1,1}$ counts *Kähler moduli* (deformations of J) and $h^{2,1}$ counts *complex structure moduli* (infinitesimal deformations of the complex structure on X , equivalently classes in $H^1(X, T_X)$).

Geometric datum	Cohomology group	Dimension
Kähler deformations	$H^{1,1}(X, \mathbb{R})$	$h^{1,1}$
Complex structure deformations	$H^1(X, T_X) \cong H^{2,1}(X)$	$h^{2,1}$
Holomorphic volume form	$H^{3,0}(X)$	1

1.7 Massless spectrum and moduli of CY compactifications

We now identify the four-dimensional moduli with cohomological data on X , following [7] § 2.5.

1.7.1 Geometric moduli: Kähler and complex structure

Proposition 1.10 (Kähler moduli). *The space of infinitesimal Ricci-flat Kähler deformations of (X, g) at fixed complex structure is $H^{1,1}(X, \mathbb{R})$. Concretely, every δJ with $\delta g_{i\bar{j}} = i \delta J_{i\bar{j}}$ harmonic of type $(1,1)$ deforms the metric to a nearby Ricci-flat one.*

Proposition 1.11 (Complex structure moduli; Kodaira–Spencer). *The space of infinitesimal complex structure deformations of X is $H^1(X, T_X) \cong H^{2,1}(X)$, where the isomorphism is given by contraction with the holomorphic volume form Ω ,*

$$\mu \mapsto \iota_\mu \Omega \in H^{2,1}(X), \quad (\iota_\mu \Omega)_{ij\bar{k}} = \Omega_{ij\bar{l}} \mu^l_{\bar{k}}, \quad (1.14)$$

where $\mu \in H^1(X, T_X)$ is the Beltrami differential, a $(0,1)$ -form valued in $T^{1,0}X$.

We write $J = t^a D_a$, $a = 1, \dots, h^{1,1}$, for real Kähler coordinates in a chosen divisor basis, with admissibility meaning $J \in \mathcal{K}(X)$, and z^i , $i = 1, \dots, h^{2,1}$, for complex parameters on $H^{2,1}(X)$. These are the precise geometric realisations of the schematic parameters $u^i(x)$ in §1.2: they are infinitesimal metric deformations δg for which the deformed metric remains Ricci-flat to first order. Yau’s theorem upgrades these infinitesimal deformations to genuine nearby Ricci-flat Kähler metrics inside the Kähler cone and complex structure moduli space.

1.7.2 How ten-dimensional fields become four-dimensional fields

We can now turn to the idea of Kaluza–Klein reductions. Suppose first that $A = 0$ and that the internal space X is fixed. A ten-dimensional scalar field admits a formal expansion in eigenfunctions of the scalar Laplacian on X ,

$$\Phi(x, y) = \sum_i \varphi_i(x) f_i(y), \quad -\Delta_X f_i = \lambda_i f_i. \quad (1.15)$$

After inserting this expansion into the ten-dimensional kinetic term and integrating over X , the coefficient $\varphi_i(x)$ becomes a four-dimensional scalar field with mass squared proportional to λ_i , up to the conventional length scale of X and to corrections from warping or background fluxes. Thus harmonic modes, $\lambda_i = 0$, give massless fields before additional interactions are included, while non-zero eigenvalues give the Kaluza–Klein tower. A field is called *light* if its mass lies below the cutoff of the four-dimensional effective theory under consideration.

The same idea applies to differential forms. If C_p is a p -form field and ω_α is a harmonic p -form on X , then

$$C_p(x, y) = \sum_\alpha a_\alpha(x) \omega_\alpha(y) + \dots, \quad [\omega_\alpha] \in H^p(X, \mathbb{R}), \quad (1.16)$$

produces scalar fields $a_\alpha(x)$ in four dimensions. This is the first reason cohomology appears in the physics: it organises the massless topological modes.

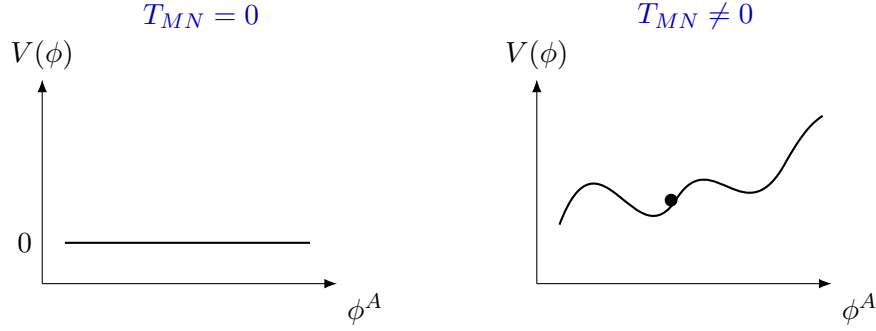


Figure 2: Classically the Calabi–Yau moduli are flat directions: in the leading source-free approximation the ten-dimensional stress-energy contribution is $T_{MN} = 0$, and the four-dimensional potential vanishes along the moduli space. Fluxes, localised sources, and further corrections are represented schematically by $T_{MN} \neq 0$ in the ten-dimensional equations; in the four-dimensional theory they lift the flat moduli space to an energy landscape $V(\phi)$.

1.8 Energy densities from moduli fields

The previous subsection explained how cohomology classes and families of internal metrics give rise to light four-dimensional fields. The next question is dynamical: once these fields have been identified, what energy is assigned to a configuration of them? This is the point at which the geometry of the compact space becomes an input to four-dimensional cosmology.

For the physics, the crucial extra step is to allow these parameters to depend on the four-dimensional coordinates x^μ . A path in the moduli space of Ricci-flat metrics then becomes a four-dimensional field configuration. If the fields vary in time, the internal metric $g^{(6)}(y; \phi(t))$ also changes in time, and this time-dependence contributes to the four-dimensional stress-energy. In this sense, the variation of the geometry of X leads directly to cosmological evolution in $3 + 1$ dimensions.

The phrase *cosmological evolution* will be used here in a minimal sense. It means a solution of the four-dimensional Einstein equations coupled to scalar fields, typically with a time-dependent metric on $\mathcal{M}_4$ and time-dependent scalar fields $\phi^A(t)$. From the ten-dimensional point of view, this same solution can be read as a family of internal geometries $g^{(6)}(y; \phi(t))$ evolving over four-dimensional time. Thus the dynamical evolution of the internal metric is not an extra metaphor: it is precisely the four-dimensional dynamics of the moduli fields.

This brings the discussion back to the central aim of the lecture series. Moving in the moduli space of internal geometries is, in the four-dimensional effective theory, the same as changing the values of scalar fields. The scalar potential on this field space is the object whose critical values are the vacuum energies that Lectures 2 and 3 compute from toric data.

The metric itself also reduces. A family of Ricci-flat metrics $g^{(6)}(y; \phi)$ depending on parameters ϕ^A gives four-dimensional scalar fields by allowing those parameters to vary over spacetime,

$$g_{mn}^{(6)}(y; \phi) \rightsquigarrow g_{mn}^{(6)}(y; \phi(x)). \quad (1.17)$$

If varying ϕ^A preserves the leading-order ten-dimensional equations, then ϕ^A is a *modulus*. For Calabi–Yau compactifications, the most important examples are the Kähler moduli and complex structure moduli counted by $h^{1,1}(X)$ and $h^{2,1}(X)$. They are geometric fields in four dimensions: their values determine the shape and size data of the internal Calabi–Yau metric. Classically, these moduli have no potential: moving in the moduli space still gives a vacuum solution, so

$$V(\phi) = 0 \quad \text{along the classical Calabi–Yau moduli space.} \quad (1.18)$$

Later we will include ingredients with non-trivial stress-energy, such as fluxes, branes, orientifold planes, and quantum corrections. In ten dimensions these appear as $T_{MN} \neq 0$ in (1.9); in

the four-dimensional effective theory they generate a scalar potential $V(\phi)$ for the fields ϕ^A . At the level of the resulting four-dimensional effective action one has schematically

$$S_4 = \int_{\mathcal{M}_4} d^4x \sqrt{-g_4} \left[\frac{M_{\text{Pl}}^2}{2} R^{(4)} - G_{AB}(\phi) \partial_\mu \phi^A \partial^\mu \phi^B - V(\phi) + \dots \right]. \quad (1.19)$$

For constant scalar fields, the scalar equations of motion reduce to

$$\partial_A V(\phi_*) = 0. \quad (1.20)$$

The four-dimensional Einstein equation then relates the value of the potential to the spacetime curvature,

$$R_{\mu\nu}^{(4)} = \frac{V(\phi_*)}{M_{\text{Pl}}^2} g_{\mu\nu}^{(4)} \quad \text{in these conventions.} \quad (1.21)$$

This is the energy-landscape picture used throughout the lectures: geometry supplies fields and functions; a vacuum is a critical point; and its vacuum energy is the critical value $V(\phi_*)$. Lecture 2 identifies the geometric and arithmetic data needed to write the four-dimensional functions K , W , and hence V ; Lecture 3 then uses these functions to compute $V(\phi_*)$ at explicit candidate critical points.

1.9 Why Calabi–Yau hypersurfaces in toric varieties?

The logic above tells us which manifolds are admissible; it does not tell us which ones we can compute on. The practical compactification programme is to enumerate large families of Calabi–Yau threefold topologies, compute the cohomological and intersection-theoretic data controlling the four-dimensional fields, compute periods and instanton corrections where needed, and then use these inputs to study potentials and vacuum energies. For computability we use the toric construction of Batyrev [5], with which the audience is already familiar. Projective hypersurfaces such as the quintic are familiar first examples; Batyrev hypersurfaces in toric varieties are the high-dimensional computational generalisation. A one-paragraph summary, for the sake of fixing notation:

Theorem 1.12 (Batyrev; CY threefolds from reflexive polytopes). *Let $\Delta^\circ \subset N_{\mathbb{R}}$ be a reflexive 4-polytope (so $\Delta = (\Delta^\circ)^\vee \subset M_{\mathbb{R}}$ is also a lattice polytope, and Δ° has the origin as its unique interior lattice point). Let $\Sigma_{\mathcal{T}}$ be the fan of a fine, regular, star triangulation $\mathcal{T}$ of Δ° . Then $V_{\Sigma_{\mathcal{T}}}$ is an MPCP toric ambient variety: it is allowed to have terminal quotient singularities, but its singular locus does not meet a generic anticanonical hypersurface in the threefold case. Consequently the generic anticanonical hypersurface $X = \{f = 0\} \subset V_{\Sigma_{\mathcal{T}}}$ is a smooth Calabi–Yau threefold, and Batyrev gives explicit combinatorial formulas for its Hodge numbers in terms of Δ, Δ° alone.*

The Kreuzer–Skarke list [6] of all Δ° contains 473 800 776 reflexive 4-polytopes; together with the FRSTs $\mathcal{T}$ this gives a landscape of explicit Calabi–Yau hypersurfaces in toric varieties covering $1 \leq h^{1,1} \leq 491$. This is the input to Lectures 2 and 3: every concrete construction will be “on a particular Batyrev hypersurface in a specified toric ambient.”

Here is the toric punchline in the form needed for the rest of the lectures. The Batyrev input $(\Delta^\circ, \mathcal{T})$ is not the final object of interest. It is the starting point for a chain of computations

$$(\Delta^\circ, \mathcal{T}) \rightsquigarrow X \rightsquigarrow \{\text{topological data, cones, periods, etc.}\} \rightsquigarrow (K, W, V) \rightsquigarrow V(\phi_*). \quad (1.22)$$

The last number is the vacuum energy of a four-dimensional critical point. The list below is therefore a dictionary, not a collection of unrelated toric facts. For each entry one should keep three questions separate: what is the geometric object, where does it enter the four-dimensional calculation, and how is it obtained from the toric description?

- *Wall data.* A torus-invariant divisor $\widehat{D}_\rho$ on the ambient toric variety restricts to a divisor $D_\rho = \widehat{D}_\rho \cap X$ on X , possibly empty for lattice points interior to facets. In favourable cases these restricted toric divisor classes span $H^{1,1}(X)$. Choosing a basis D_a , one computes

$$\kappa_{abc} = \int_X D_a \wedge D_b \wedge D_c, \quad c_{2,a} = \int_X c_2(X) \wedge D_a. \quad (1.23)$$

Together with the Hodge numbers $h^{1,1}, h^{2,1}$, this completes the *Wall data* of a Calabi–Yau threefold. For $J = t^a D_a$, these same numbers give the Calabi–Yau volume and divisor volumes

$$\begin{aligned} \mathcal{V}(X, J) &= \frac{1}{3!} \int_X J \wedge J \wedge J = \frac{1}{6} \kappa_{abc} t^a t^b t^c, \\ \tau_a &= \frac{1}{2} \int_{D_a} J \wedge J = \frac{1}{2} \int_X D_a \wedge J \wedge J = \frac{\partial \mathcal{V}}{\partial t^a} = \frac{1}{2} \kappa_{abc} t^b t^c. \end{aligned} \quad (1.24)$$

Used in: the Kähler potential, the mirror prepotential, and the non-perturbative superpotential.

How: In practice this data is obtained from the ambient toric intersection ring, the hypersurface class, and the Chern-class formula for toric hypersurfaces. The implementation used in the computational parts of these lectures is **CYTools** [10]; the same data are the Wall data used to compare toric phases in [18].

- *Cone data and flops.* The Kähler cone $\mathcal{K}(X)$ records which real $(1,1)$ -classes can be represented by Kähler forms; its dual Mori cone records effective curve classes. Curve volumes are the inequalities that keep the large-volume expansion under control. Different FRSTs can give adjacent Kähler chambers related by flops, and the union of these chambers is the extended Kähler cone $\widetilde{\mathcal{K}}(X)$. In Lecture 3, moving through these adjacent chambers is the geometric part of the vacuum search: one follows the potential while checking that all relevant curve volumes remain positive and sufficiently large.

Used in: the domain of validity of the large-volume expansion, the definition of Kähler chambers, and the geometric transitions used in the Lecture 3 search strategy.

How: **CYTools** computes Mori and Kähler cone data for toric phases and tracks the classical data across triangulations [10]. The secondary-fan and chamber-navigation layer used in the separate Lecture 3 and 4 discussion is provided by **fanroots** [19].

- *Divisor data.* Non-perturbative terms in the superpotential may come from ED3 instantons or gaugino condensation on divisors. A necessary zero-mode test is expressed in terms of the Hodge numbers of the divisor, especially $h^{0,1}(D)$ and $h^{0,2}(D)$, or equivalently the arithmetic genus appearing in Witten’s criterion. For prime toric divisors these Hodge numbers can often be read from the face of Δ dual to the lattice point defining the divisor. This is why a purely combinatorial computation can screen candidate instanton divisors before any four-dimensional potential is written.

Used in: deciding which divisors can support D-instanton terms in W_{np} .

How: For prime toric divisors, compute divisor Hodge numbers via the face-duality formulae, see e.g. [20]. This is a special toric shortcut. For non-inherited or non-toric divisors, determining the relevant cohomology groups is a substantially harder sheaf-cohomology problem and should not be treated as automatic output of the toric combinatorics.

- *Period data.* The flux superpotential in Type IIB is built from periods of the holomorphic three-form,

$$\Pi(z) = \left(\int_{\Gamma_I} \Omega(z) \right), \quad \Gamma_I \in H_3(X, \mathbb{Z}). \quad (1.25)$$

Near a large complex structure point these periods are computed by mirror symmetry. In practice this means solving the Picard–Fuchs system of the mirror polytope and expressing the answer in flat coordinates. These periods determine W_{flux} , and hence the flux value W_0 after the complex structure moduli and the axio-dilaton are stabilised.

Used in: the flux superpotential W_{flux} , in particular the stabilisation of complex structure moduli.

How: construct the GKZ / Picard–Fuchs system from the Batyrev mirror data, compute the Frobenius solutions near an LCS point, and then match the logarithmic solutions to an integral symplectic period basis using the large volume monodromy on the mirror side [21, 22, 23].

- *Enumerative data.* Genus-zero Gopakumar–Vafa invariants n_β , for $\beta \in H_2(X, \mathbb{Z})$, are the integer invariants that repackage genus-zero Gromov–Witten invariants after the multiple-cover contributions have been separated [24, 23]. For the audience it is useful to remember two interpretations. Mathematically, they are the integral BPS/GV refinement of rational curve counts in class β . In the physical interpretation they are indexed BPS degeneracies, naturally described by M2-branes wrapping holomorphic curves in class β . They appear in the worldsheet-instanton part of the prepotential, so they correct the periods and can generate exponentially small terms used in small- W_0 constructions. Lecture 2 is devoted to making this computational mirror symmetry (CMS) / GV calculation explicit enough for the vacuum energy story.

Used in: worldsheet-instanton corrections to the prepotential, convergence estimates for the instanton expansion, and the small terms that can enter small- W_0 and conifold-PFV constructions.

How: compare the instanton part of the mirror prepotential with the Frobenius period expansion, use the multiple-cover formula to pass from genus-zero Gromov–Witten coefficients to integral GV invariants, and organise the resulting finite computation by Mori-cone degree or by a decomposition-closed truncation [24, 21, 23]. In Lecture 3 the same curve-class data are then compared across flop-related chambers.

Many of the conceptual ingredients in this story are classical by now: Calabi–Yau compactification, mirror symmetry, periods, flux superpotentials, instanton corrections, and Gopakumar–Vafa invariants have been studied for decades. What has changed rather dramatically is our ability to compute these ingredients in practice, not only for a few hand-picked examples, but uniformly across large families with widely varying Hodge numbers. The main breakthrough for explicit string constructions is therefore algorithmic as much as conceptual: toric geometry, mirror symmetry, triangulation algorithms, period computations, and modern searches over flux and Kähler data turn the existence of a vast landscape into concrete, checkable Diophantine problems.

The companion **CYTools** session demonstrates the computational pipelines on explicit examples: querying the Kreuzer–Skarke database, constructing or sampling FRSTs, building a **CalabiYau** object, computing intersection numbers and cone data, testing divisor topology, and extracting truncated GV dictionaries. The remainder of these lectures explains why these computations are the inputs to established string theory constructions.

Lecture 1 punchlines. The first lecture should leave the following takeaways. First, compactification turns internal geometry into four-dimensional fields; the light fields are controlled by cohomology, harmonic forms, and Ricci-flat metric deformations. Second, uncorrected Calabi–Yau compactifications have flat moduli spaces, so vacuum energies only become isolated numbers after non-trivial sources of stress-energy generate a potential. Third, Calabi–Yau threefolds enter because Yau’s theorem turns the nonlinear Ricci-flat metric problem into a computable complex-geometric existence problem. Fourth, hypersurfaces à la Batyrev make this computable

at scale: toric data provide the intersection, cone, divisor, period, and GV inputs that Lectures 2 and 3 feed into the four-dimensional scalar potential.

That completes Lecture 1.

2 Lecture 2: Periods and GV invariants from toric data

Anchor: TASI [7], §§ 3–6.

Guiding question. Lecture 1 ended with a flat moduli space: the Calabi–Yau metric can move in continuous families, and at leading order the associated four-dimensional scalar fields have $V = 0$. Lecture 2 asks how one computes the first ingredient that turns this geometric moduli space into an explicit function in the four-dimensional theory. The answer is a period calculation. Quantised three-form fluxes choose an integral linear combination of periods of the holomorphic three-form, so that

$$W_{\text{flux}} = \int_X \Omega \wedge G_3. \quad (2.1)$$

The target of the lecture is therefore deliberately narrow: compute W_{flux} by computing periods. Mirror symmetry converts the period problem near a large complex structure point of X into a large volume calculation on the mirror $\tilde{X}$. For toric hypersurfaces, the relevant data are finite and combinatorial: the reflexive polytope, the GLSM charge matrix, the intersection numbers, and the genus-zero Gopakumar–Vafa invariants [25, 24] that organise the worldsheet instanton contributions to the mirror prepotential [26, 27]. The final section then uses this period machinery in one application, namely the perturbatively flat vacuum construction of [28].

The effective field theory and compactification background follow the conventions of [7], especially the material on flux compactifications and periods. The expanded discussion of period computation and genus-zero GV data below follows the computational mirror-symmetry framework of [23, 29]. Notice the order of logic: we first compute periods and the flux superpotential; only at the end do we discuss a particular flux-vacuum application.

Historical and computational thread. The bridge from Calabi–Yau geometry to numbers in the four-dimensional theory is mirror symmetry. The quintic calculation of Candelas, de la Ossa, Green and Parkes [30], Witten’s topological field theory formulation [31], the two-parameter examples [26, 27], and the Hosono–Klemm–Theisen–Yau (HKTY) period/mirror-map construction [21, 32] provide the historical route from periods on the mirror to instanton-corrected prepotentials. The SYZ and Hori–Vafa perspectives give complementary geometric and physical explanations of mirror symmetry [33, 34, 35], although the computations below use the period/GKZ side of the story. Batyrev’s toric mirror construction [5] and the large complex structure compactification picture [36] explain why this is the natural language for hypersurfaces in toric varieties. The Gopakumar–Vafa invariants [25, 24] then give the integral curve-counting data that organise the genus-zero instanton expansion used below. Computational mirror symmetry in the sense of [23] is the high-dimensional continuation of this story: it asks how to compute the period vector, mirror map and GV data reliably when the Picard rank is large and the Mori cone need not be simplicial.

2.1 What has to be computed?

The first point of the lecture is not yet mirror symmetry. It is to identify the four-dimensional quantity for which the geometry of X will become useful. A compactification turns continuous families of Calabi–Yau metrics and background fields into scalar fields in four spacetime dimensions. Locally on field space we write these fields as

$$\Phi^I = (\tau, z^a, T_i). \quad (2.2)$$

Here z^a , with $a = 1, \dots, h^{2,1}(X)$, are local coordinates on complex structure moduli space; T_i , with $i = 1, \dots, h^{1,1}(X)$, are complexified Kähler moduli; and τ is the axio-dilaton. The ten-dimensional origin of τ and of the three-form fluxes is recalled in §2.3; for the present section only the four-dimensional role of these fields matters.

For slowly varying fields, the scalar part of the four-dimensional Einstein-frame action has the schematic form

$$S_4 = \int d^4x \sqrt{-g_4} \left(\frac{1}{2} R_4 - K_{I\bar{J}}(\Phi, \bar{\Phi}) \partial_\mu \Phi^I \partial^\mu \bar{\Phi}^{\bar{J}} - V(\Phi, \bar{\Phi}) + \dots \right). \quad (2.3)$$

This is the lecture’s basic organising equation. The coefficient $K_{I\bar{J}}$ is the metric on the scalar field space, so it tells us what “gradient” means in the equations of motion. The scalar potential V is the energy density for constant fields, and its critical points are the candidate four-dimensional vacua. More explicitly, the field equations have the schematic form

$$\square \Phi^I + \Gamma_{JK}^I \partial_\mu \Phi^J \partial^\mu \Phi^K = K^{I\bar{J}} \partial_{\bar{J}} V, \quad (2.4)$$

up to the conventional sign determined by the spacetime signature. For constant scalar fields this reduces to the critical-point equations

$$\partial_I V = 0. \quad (2.5)$$

Thus the vacuum problem asks for two pieces of data: the field-space metric and the scalar potential.

The field-space metric is the first place where the geometry of the Calabi–Yau enters. In the supersymmetric compactifications used in these notes, $K_{I\bar{J}}$ is a Kähler metric,

$$K_{I\bar{J}} = \partial_I \partial_{\bar{J}} K, \quad (2.6)$$

for a real function K , the Kähler potential. On the complex structure moduli space this metric is the Weil–Petersson metric. The Hodge-theoretic formula is

$$K_{\text{cs}} = -\log \left(-i \int_X \Omega \wedge \bar{\Omega} \right), \quad G_{ab}^{\text{cs}} = \partial_a \partial_{\bar{b}} K_{\text{cs}}. \quad (2.7)$$

This single display contains the reason periods are unavoidable: $\int_X \Omega \wedge \bar{\Omega}$ is computed from the periods of the holomorphic three-form. Consequently, periods determine not only a holomorphic superpotential later on, but also the metric and connection used in the equations of motion.

The Kähler-moduli metric is given by

$$G_{ij}^K = \frac{1}{4\mathcal{V}(X, J)} \int_X \omega_i \wedge \star_6 \omega_j, \quad (2.8)$$

for a basis $\omega_i \in H^{1,1}(X)$, $i = 1, \dots, h^{1,1}(X)$. As shown in §2.2, it is controlled by the classical volume function. If

$$J = t^i \omega_i \quad (2.9)$$

is the Kähler form expanded in this basis, then

$$\mathcal{V}(X, J) = \frac{1}{3!} \int_X J \wedge J \wedge J = \frac{1}{6} \kappa_{ijk} t^i t^j t^k, \quad \kappa_{ijk} = \int_X \omega_i \wedge \omega_j \wedge \omega_k. \quad (2.10)$$

The real parts of the complexified Kähler coordinates used below are divisor volumes:

$$\tau_i = \text{Re}(T_i) = \frac{1}{2} \int_X \omega_i \wedge J \wedge J = \frac{\partial \mathcal{V}}{\partial t^i} = \frac{1}{2} \kappa_{ijk} t^j t^k. \quad (2.11)$$

At leading order in α' and g_s , the Kähler potential used for the compactification examples is therefore

$$K(\tau, z, T; \bar{\tau}, \bar{z}, \bar{T}) = -\log(-i(\tau - \bar{\tau})) - \log\left(-i \int_X \Omega \wedge \bar{\Omega}\right) - 2 \log \mathcal{V}(T, \bar{T}). \quad (2.12)$$

The first term is the axio-dilaton metric, the second is the Weil–Petersson metric on complex structure moduli space, and the third is the large-volume Kähler-moduli metric. The more detailed metric interpretation is given in §2.2; the important message here is that each term in K is computable from geometric data.

We can now introduce the scalar potential in the form used later. In four-dimensional $\mathcal{N} = 1$ supergravity, the F -term potential is

$$V = e^K \left(K^{I\bar{J}} D_I W \overline{D_{\bar{J}} W} - 3|W|^2 \right), \quad D_I W = \partial_I W + (\partial_I K) W. \quad (2.13)$$

Here W is a holomorphic superpotential. Thus the compactification problem becomes concrete only after we can compute K , W , and the region in which the formulae are reliable. Lecture 2 focuses on the part of this problem that toric mirror symmetry computes most directly: the flux superpotential

$$W_{\text{flux}}(\tau, z) = \int_X \Omega(z) \wedge G_3(\tau). \quad (2.14)$$

The same periods of Ω that occur in K_{cs} also occur in this integral. Quantised fluxes choose an integral linear combination of those periods. Therefore the computational target for the rest of the lecture is

$$\text{toric data} \implies \text{period vector of } \Omega \implies K_{\text{cs}} \text{ and } W_{\text{flux}}, \quad (2.15)$$

which we then use to find explicit solutions to

$$D_{\tau} W_{\text{flux}} = D_a W_{\text{flux}} = 0. \quad (2.16)$$

For generic fluxes, the last equations are transcendental and are solved numerically. The final section of the lecture explains why perturbatively flat vacua provide a special class in which the polynomial large complex structure part can be solved analytically, and the GV-controlled instanton terms then provide the first non-trivial lifting.

2.2 Aside: Weil–Petersson metrics on moduli spaces of Calabi–Yau threefolds*

The four-dimensional $\mathcal{N} = 1$ effective theory associated with a flux compactification on a Calabi–Yau threefold X inherits two distinct metrics on its scalar manifold: one on the complex structure moduli space $\mathcal{M}_{\text{cs}}(X)$, the other on the complexified Kähler moduli space $\mathcal{M}_{\text{K}}(X)$. Both are Weil–Petersson metrics in the classical differential-geometric sense, and both admit two complementary descriptions:

- an L^2 -pairing of metric variations of the underlying Ricci-flat metric on X ;
- a Hodge-theoretic or intersection-theoretic description in terms of the holomorphic three-form Ω on the complex structure side, or the Kähler form J on the Kähler side.

The point of the second description is computational: it replaces an unknown Ricci-flat metric by periods and intersection numbers. The unobstructedness of complex structure deformations is the Tian–Todorov theorem [37, 38]; the equality between the metric-variation pairing and the Hodge-theoretic formula below is the standard variation of Hodge structure description of the Weil–Petersson metric.

Complex structure moduli space $\mathcal{M}_{\text{cs}}(X)$

Let $g_{m\bar{n}}^{(0)}$ denote the Ricci-flat Kähler metric on X supplied by Yau’s theorem. Complex structure deformations, parametrised by local coordinates z^a with $a = 1, \dots, h^{2,1}(X)$, appear as variations of the metric of pure type $(2, 0)$ and $(0, 2)$. The natural L^2 inner product of such variations is the Weil–Petersson metric [39, 40]

$$2 G_{a\bar{b}}^{\text{cs}} dz^a d\bar{z}^{\bar{b}} = \frac{1}{2\mathcal{V}} \int_X g^{m\bar{n}} g^{k\bar{l}} \delta g_{mk} \delta g_{\bar{n}\bar{l}} \sqrt{g} d^6 y. \quad (2.17)$$

Here $\mathcal{V} = \int_X \sqrt{g} d^6 y$ is the total volume of X . The factor of 2 on the left records the convention $ds^2 = 2 G_{a\bar{b}} dz^a d\bar{z}^{\bar{b}}$; normalisations differ slightly in the literature.

Via the Calabi–Yau condition, each complex structure deformation corresponds to a harmonic $(2, 1)$ -form. Concretely, the variation of Ω satisfies Griffiths transversality,

$$\partial_a \Omega = k_a(z) \Omega + \chi_a, \quad \chi_a \in H^{2,1}(X). \quad (2.18)$$

Equivalently, after choosing the normalisation determined by K_{cs} below,

$$\chi_a = D_a \Omega = \partial_a \Omega + (\partial_a K_{\text{cs}}) \Omega. \quad (2.19)$$

The Hodge-theoretic form of the Weil–Petersson metric is

$$G_{a\bar{b}}^{\text{cs}} = - \frac{\int_X \chi_a \wedge \bar{\chi}_{\bar{b}}}{\int_X \Omega \wedge \bar{\Omega}}. \quad (2.20)$$

The denominator is the natural Hermitian metric on the Hodge line bundle whose fibre is $H^{3,0}(X)$. In particular,

$$K_{\text{cs}} = -\log\left(-i \int_X \Omega \wedge \bar{\Omega}\right), \quad G_{a\bar{b}}^{\text{cs}} = \partial_a \partial_{\bar{b}} K_{\text{cs}}. \quad (2.21)$$

A short verification uses (2.18) and $\int_X \Omega \wedge \chi_a = 0$ to obtain

$$\partial_a \partial_{\bar{b}} K_{\text{cs}} = - \frac{\int_X \chi_a \wedge \bar{\chi}_{\bar{b}}}{\int_X \Omega \wedge \bar{\Omega}}. \quad (2.22)$$

Thus the period calculation is already a calculation of the geometry of the equations of motion: the same periods determine the holomorphic function W_{flux} and the metric and connection appearing in $D_a W_{\text{flux}}$.

Kähler moduli space $\mathcal{M}_{\text{K}}(X)$

The Kähler class $[J]$ of the Ricci-flat metric is parametrised, in a basis $\{\omega_i\}_{i=1}^{h^{1,1}(X)}$ of $H^{1,1}(X, \mathbb{Z})$, by real moduli t^i lying in the Kähler cone $\mathcal{K}(X) \subset H^{1,1}(X, \mathbb{R})$:

$$J = \sum_i t^i \omega_i \in \mathcal{K}(X). \quad (2.23)$$

The intersection numbers

$$\kappa_{ijk} = \int_X \omega_i \wedge \omega_j \wedge \omega_k \in \mathbb{Z} \quad (2.24)$$

and the Kähler cone determine the classical Kähler-side geometry.

The L^2 inner product of Kähler metric variations $\delta g_{m\bar{n}} = (\omega_i)_{m\bar{n}} \delta t^i$ induces the Weil–Petersson metric on Kähler moduli space [41, 39]

$$G_{ij}^{\text{K}} = \frac{1}{4\mathcal{V}} \int_X \omega_i \wedge \star_6 \omega_j. \quad (2.25)$$

The $1/(4\mathcal{V})$ prefactor here is fixed by the Hodge-star action on $H^{1,1}$ and the volume normalisation. Applying the Hodge star on $(1,1)$ -forms and using the hard-Lefschetz decomposition gives the closed expression

$$G_{ij}^K = -\frac{\kappa_{ijk}t^k}{4\mathcal{V}} + \frac{\mathcal{T}_i\mathcal{T}_j}{4\mathcal{V}^2}, \quad \mathcal{T}_i = \frac{1}{2} \int_X J \wedge J \wedge \omega_i = \frac{1}{2} \kappa_{ijk} t^j t^k. \quad (2.26)$$

Here $\mathcal{T}_i$ is the volume of the divisor dual to ω_i ; in the main flow it is denoted $\tau_i = \text{Re } T_i$ after choosing the corresponding divisor basis. The corresponding tree-level Kähler potential is

$$K_{\text{tree}} = -2 \log \mathcal{V} = -2 \log \left(\frac{1}{6} \kappa_{ijk} t^i t^j t^k \right). \quad (2.27)$$

Differentiating K_{tree} with respect to the complex chiral coordinates T_i reproduces (2.26), up to the Jacobian factors from the change of variables $t^i \mapsto T_i$.

2.3 Aside: Minimal Type IIB and orientifold dictionary*

Only a small part of the ten-dimensional Type IIB vocabulary is needed for the computations in this lecture. The bosonic fields that enter the formulae below are

$$g_{MN}, \quad \tau = C_0 + i e^{-\phi}, \quad B_2, C_2, C_4. \quad (2.28)$$

Here g_{MN} is the ten-dimensional metric, ϕ is the dilaton, C_0 is the RR axion, B_2 and C_2 are two-form gauge potentials, and C_4 is a four-form gauge potential. Their field strengths are

$$H_3 = dB_2, \quad F_3 = dC_2, \quad G_3 = F_3 - \tau H_3, \quad (2.29)$$

with the modified self-dual five-form built from C_4 not needed explicitly in this lecture. Thus the symbol τ in the four-dimensional theory is the zero mode of the axio-dilaton, while F_3 and H_3 are the quantised three-form fluxes whose periods enter W_{flux} .

Definition 2.1 (O3/O7 orientifold of X). *An O3/O7 orientifold of X is the quotient by a holomorphic involution $\sigma : X \rightarrow X$ with $\sigma^2 = \text{id}$ and $\sigma^*\Omega = -\Omega$, combined in the physical theory with worldsheet orientation reversal and fermion parity. The fixed locus of σ is a disjoint union of isolated points, called O3-planes, and holomorphic divisors, called O7-planes. The induced splitting*

$$H^{p,q}(X) = H_+^{p,q}(X) \oplus H_-^{p,q}(X) \quad (2.30)$$

selects the fields kept by the orientifold projection: in the O3/O7 setting used here, the Kähler moduli are counted by $h_+^{1,1}$, and the complex structure moduli in the flux superpotential by $h_-^{2,1}$. Sign conventions for σ^Ω differ in the literature; the convention displayed above matches the one used for the flux formulae in these notes and is the convention of [42], while [43] uses the opposite sign convention.*

For the toric hypersurfaces used in these lectures there is a particularly effective way of finding orientifold involutions that are inherited from the ambient toric variety. Let $X = \{f = 0\} \subset V_\Sigma$ be a Batyrev hypersurface, with V_Σ described by Cox coordinates x_p indexed by the rays of Σ , hence by non-zero lattice points $p \in \Delta^\circ \cap N$ in an MPCP subdivision. The method of [44] first replaces the problem of finding holomorphic involutions of X by the combinatorial problem of finding involutions of the ambient Cox quotient. After gauge fixing in the ambient automorphism group, the representatives used in this enumeration have the form

$$\tilde{\sigma} = \phi_{[t]} \circ L, \quad (2.31)$$

where L is a lattice automorphism of Δ° preserving the relevant fan or regular subdivision, and $\phi_{[t]}$ is the torus automorphism determined, after gauge fixing, by a half-integral class $[t] \in \frac{1}{2}N/N$.

The condition $\tilde{\sigma}^2 = 1$ and the residual conjugacy action give a finite list of representatives. One then asks whether the locus of L -symmetric height functions meets the secondary fan in a fine regular star triangulation, or at least in a fine regular star subdivision whose restrictions to all two-faces of Δ° are fine and regular. This is the toric condition under which the invariant ambient toric variety may be singular in the symmetric limit while the generic invariant hypersurface remains smooth.

The hypersurface equation is not arbitrary after the involution has been chosen. Writing

$$f = \sum_{q \in \Delta \cap M} \psi_q s_q, \quad (2.32)$$

one imposes covariance

$$\tilde{\sigma}^* f = \gamma_f f \quad (2.33)$$

for a fixed sign character γ_f . Equivalently, the coefficients ψ_q are constrained along L -orbits, with phases determined by t . The sign of the induced action on the Poincaré residue of Ω then distinguishes the O3/O7 case from the O5/O9 case. The fixed locus is computed directly from the toric data. If Σ is the chosen invariant fan, its relevant irreducible components are labelled by pairs (ν, σ) with

$$\sigma \in \Sigma \text{ pointwise fixed by } L, \quad \nu \in H_L^-, \quad t + \nu + \frac{1}{2} \sum_{p \in \sigma(1)} p \in N, \quad (2.34)$$

where $H_L^- = P_L^-(N)/(2P_L^-(N))$ and $P_L^- = (1 - L)/2$ records the inequivalent anti-invariant toric phases; in the displayed integrality condition t and ν denote chosen representatives of their classes. One then removes components contained in larger fixed components. These ambient fixed components are intersected with the invariant hypersurface. Complex codimension-one components on X give O7 divisors and complex codimension-three components give O3 points.

This construction also computes the orientifold Hodge splitting. In the favourable case, the induced permutation of toric divisor classes gives the $\pm$ eigenspaces of $H^{1,1}(X)$; the remaining split of complex structure moduli is then obtained from the Lefschetz fixed-point formula, using the Euler characteristic of the fixed locus. In the non-favourable case one must enlarge this divisor calculation by the irreducible components of toric divisors that split when restricted to X , notably those coming from points interior to two-faces of Δ° . This is the “orientifold weighted Hodge number” computation of [44]. For the present lecture, the important point is that the orientifold data become finite combinatorial input read from the same toric description that also supplies the periods, intersection numbers, and curve data.

For the rest of the lecture we mostly work with orientifold examples satisfying

$$h_-^{1,1} = 0, \quad h_+^{2,1} = 0. \quad (2.35)$$

Then $h_+^{1,1} = h^{1,1}$ and $h_-^{2,1} = h^{2,1}$. From this point on we suppress the $\pm$ labels on the surviving cohomology groups and Hodge numbers, unless the orientifold parity itself is the point under discussion. In particular, the flux lattice will be written simply as $H^3(X, \mathbb{Z})$ in the formulae below.

2.4 Flux quantisation, periods, and the GVW input

We now isolate the computational input that produces W_{flux} . A flux configuration is not a real vector of continuous parameters. It is a pair of integral cohomology classes

$$[F_3] \in H^3(X, \mathbb{Z}), \quad [H_3] \in H^3(X, \mathbb{Z}). \quad (2.36)$$

The orientifold projection has already selected the relevant surviving lattice; by the convention fixed above we do not write parity labels in the rest of the lecture. Integrality is the usual Dirac

quantisation condition: in string units, the periods over integral three-cycles are integers. Large gauge transformations of the underlying potentials C_2 and B_2 identify potentials whose periods differ by integers, so the cohomology class is the invariant datum.

Choose a symplectic basis $\{\alpha_A, \beta^A\}$ of the relevant integral cohomology lattice, with $A = 0, \dots, h^{2,1}$ in the unprojected notation and with the corresponding surviving range after the orientifold projection. The pairing is

$$\int_X \alpha_A \wedge \beta^B = \delta_A^B, \quad \int_X \alpha_A \wedge \alpha_B = \int_X \beta^A \wedge \beta^B = 0. \quad (2.37)$$

The periods of the holomorphic three-form are the coefficients in

$$\Omega(z) = X^A(z) \alpha_A - \mathcal{F}_A(z) \beta^A. \quad (2.38)$$

Near a generic point of $\mathcal{M}_{\text{cs}}(X)$, $h^{3,0}(X) = 1$ and Griffiths transversality make the X^A projective coordinates on complex structure moduli space, so $\mathcal{F}_A = \partial \mathcal{F} / \partial X^A$ for a homogeneous degree-two prepotential $\mathcal{F}$ [30]. The same period vector also gives the complex structure Kähler potential,

$$K_{\text{cs}} = -\log\left(-i \int_X \Omega \wedge \bar{\Omega}\right) = -\log\left(i(X^A \bar{\mathcal{F}}_A - \bar{X}^A \mathcal{F}_A)\right). \quad (2.39)$$

Thus computing periods is already enough to compute the metric part of the flux equations of motion.

The physical flux combination is

$$G_3 = F_3 - \tau H_3, \quad (2.40)$$

where τ is the axio-dilaton introduced in (2.28). The flux contribution to the four-dimensional superpotential is the Gukov–Vafa–Witten period pairing [45],

$$W_{\text{flux}}(\tau, z) = \int_X \Omega(z) \wedge G_3(\tau). \quad (2.41)$$

Writing

$$F_3 = f^A \alpha_A - f_A \beta^A, \quad H_3 = h^A \alpha_A - h_A \beta^A, \quad f^A, f_A, h^A, h_A \in \mathbb{Z}, \quad (2.42)$$

one obtains, up to the sign convention for the symplectic pairing,

$$W_{\text{flux}} = (f^A - \tau h^A) \mathcal{F}_A(z) - (f_A - \tau h_A) X^A(z). \quad (2.43)$$

This is the main reduction. A choice of flux is an integer vector, and the superpotential is the corresponding integer linear combination of period functions. Therefore the next mathematical task is not to solve for vacua immediately, but to compute the period vector.

At large complex structure, the period vector splits into a polynomial part and exponentially suppressed corrections.² Mirror symmetry computes these corrections from the large volume prepotential on $\tilde{X}$. The coefficients of the instanton part of that prepotential are genus-zero GV invariants. This is the precise reason the enumerative calculation below enters the lecture: it supplies the instanton-corrected period entries used in (2.43).

The supersymmetric flux equations for the axio-dilaton and complex structure moduli are

$$D_\tau W_{\text{flux}} = 0, \quad D_a W_{\text{flux}} = 0. \quad (2.44)$$

The ten-dimensional GKP interpretation of these equations is important background, but it is not needed for the period computation itself; see [42] and [7], § 3.3.2.

²A *large complex structure* (LCS) point is a boundary point of $\mathcal{M}_{\text{cs}}(X)$ at which the monodromy of the periods of Ω is maximally unipotent. Under mirror symmetry it corresponds to the large volume limit on the mirror $\tilde{X}$.

2.5 Computational mirror symmetry and Gopakumar–Vafa invariants

We have expressed W_{flux} as an integral linear combination of periods. The next task is therefore to compute those periods. For Calabi–Yau hypersurfaces in toric varieties this is most efficiently done by mirror symmetry. Periods on the Type IIB Calabi–Yau X are matched to the large volume A -model prepotential on the mirror $\tilde{X}$. The exponentially small terms in that prepotential are organised by genus-zero Gopakumar–Vafa invariants of curve classes on $\tilde{X}$ [25, 24]. The computational mirror-symmetry programme of [23, 29] turns this statement into an algorithm that remains practical for large Picard rank and for non-simplicial Mori cones.

For this workshop audience it is useful to separate the conceptual content from the algorithmic advance. The conceptual framework is the classical one: mirror symmetry near large complex structure [30, 21], Batyrev’s toric construction [5], and the GV integral reorganisation of Gromov–Witten theory [25, 24]. The newer contribution of [23] is algorithmic: it gives a practical way to compute the truncated instanton expansion when $h^{1,1}$ is large and the relevant cones are high-dimensional.

Following the flow of [23], but using the notation of the present lectures, the lecture-facing story is:

1. compute the periods of Ω near an LCS point of $\mathcal{M}_{\text{cs}}(X)$;
2. identify the same period vector, by mirror symmetry, with the Type IIA large volume period vector on $\tilde{X}$;
3. read off the instanton part of the A -model prepotential;
4. use the multiple-cover relation to extract the genus-zero GV integers;
5. use toric GKZ/Frobenius series to carry this out without constructing a global Picard–Fuchs basis.

Remark 2.2 (Notation comparison with related references). Throughout these lectures X denotes the Calabi–Yau threefold used for the Type IIB compactification, and $\tilde{X}$ denotes its mirror. Thus Type IIA in the mirror symmetry comparison lives on $\tilde{X}$. This is the convention used in the TASI discussion of computational mirror symmetry [7], §5.5.2.2. Reference [23] uses the opposite labelling in parts of its exposition; the formulae below have been translated into the notation of these lectures (hopefully consistently).

The A -model prepotential and period vector

Fix the mirror Calabi–Yau threefold $\tilde{X}$. Choose integral curve classes $\{\tilde{\gamma}^a\} \subset H_2(\tilde{X}, \mathbb{Z})$, and choose a dual basis $\{\tilde{\omega}_a\} \subset H^{1,1}(\tilde{X})$ satisfying

$$\int_{\tilde{\gamma}^a} \tilde{\omega}_b = \delta_b^a. \quad (2.45)$$

We introduce complex coordinates

$$\mathfrak{t}^a = \int_{\tilde{\gamma}^a} (\tilde{B} + i\tilde{J}) = b^a + it^a. \quad (2.46)$$

Here $\tilde{J}$ is the Kähler form on $\tilde{X}$ and $\tilde{B}$ is the mirror-side B -field. This makes $\mathfrak{t}^a$ the Type IIA complexified Kähler parameters on $\tilde{X}$.

In the large volume chamber, the genus-zero A -model prepotential has the form

$$F_0(\mathfrak{t}) = F_{0,\text{poly}}(\mathfrak{t}) + F_{0,\text{inst}}(\mathfrak{t}). \quad (2.47)$$

The polynomial part is fixed by classical topology: the triple intersections and second-Chern-class integrals

$$\tilde{\kappa}_{abc} = \int_{\tilde{X}} \tilde{\omega}_a \wedge \tilde{\omega}_b \wedge \tilde{\omega}_c, \quad \tilde{c}_a = \int_{\tilde{X}} c_2(\tilde{X}) \wedge \tilde{\omega}_a, \quad (2.48)$$

together with the Euler characteristic $\chi(\tilde{X})$. In the conventions of [23], translated to the present notation, this polynomial part is

$$F_{0,\text{poly}}(\mathbf{t}) = -\frac{1}{6}\tilde{\kappa}_{abc}\mathbf{t}^a\mathbf{t}^b\mathbf{t}^c + \frac{1}{2}\tilde{a}_{ab}\mathbf{t}^a\mathbf{t}^b + \frac{1}{24}\tilde{c}_a\mathbf{t}^a + \frac{\zeta(3)\chi(\tilde{X})}{2(2\pi i)^3}. \quad (2.49)$$

The quadratic coefficients $\tilde{a}_{ab}$ record the chosen integral symplectic basis. In the convention used by [23] and [7], §5.5.2, they are taken to be

$$\tilde{a}_{ab} = \frac{1}{2} \begin{cases} \tilde{\kappa}_{aab}, & a \geq b, \\ \tilde{\kappa}_{abb}, & a < b. \end{cases} \quad (2.50)$$

Polynomial signs and lower-degree terms vary with period-vector conventions; (2.49) is the convention used for the period matching below.

Let $\mathcal{M}(\tilde{X})$ denote the Mori cone, i.e. the cone of effective curve classes on $\tilde{X}$. In the chosen curve basis, a class is written $\tilde{\mathbf{q}} = \tilde{q}_a \tilde{\gamma}^a$, and $\tilde{\mathbf{q}} \cdot \mathbf{t}$ means $\tilde{q}_a \mathbf{t}^a$. The instanton part is most transparently written first in terms of the genus-zero Gromov–Witten invariants of $\tilde{X}$:

$$F_{0,\text{inst}}(\mathbf{t}) = -\frac{1}{(2\pi i)^3} \sum_{0 \neq \tilde{\mathbf{q}} \in \mathcal{M}(\tilde{X}) \cap H_2(\tilde{X}, \mathbb{Z})} N_{\tilde{\mathbf{q}}} q^{\tilde{\mathbf{q}}}, \quad q^{\tilde{\mathbf{q}}} = e^{2\pi i \tilde{\mathbf{q}} \cdot \mathbf{t}}. \quad (2.51)$$

Here $N_{\tilde{\mathbf{q}}} \in \mathbb{Q}$ are genus-zero Gromov–Witten invariants. The GV integers are the coefficients that repackage this same series into the polylogarithmic form

$$F_{0,\text{inst}}(\mathbf{t}) = -\frac{1}{(2\pi i)^3} \sum_{0 \neq \tilde{\mathbf{q}} \in \mathcal{M}(\tilde{X}) \cap H_2(\tilde{X}, \mathbb{Z})} n_{\tilde{\mathbf{q}}} \text{Li}_3(e^{2\pi i \tilde{\mathbf{q}} \cdot \mathbf{t}}). \quad (2.52)$$

This is the exact location at which the integers enter the period calculation: they are the coefficients of the exponentially suppressed part of the mirror prepotential, and hence of the corresponding period vector.

Before continuing with the toric computation, we spell out what these integers mean. Given genus-zero Gromov–Witten invariants $N_{\tilde{\mathbf{q}}}$, the genus-zero GV coefficients are defined by the multiple-cover relation

$$\sum_{0 \neq \tilde{\mathbf{q}}} N_{\tilde{\mathbf{q}}} q^{\tilde{\mathbf{q}}} = \sum_{0 \neq \tilde{\mathbf{q}}} n_{\tilde{\mathbf{q}}} \sum_{k \geq 1} \frac{1}{k^3} q^{k\tilde{\mathbf{q}}}. \quad (2.53)$$

Equivalently, $N_{\tilde{\mathbf{q}}} = \sum_{k|\tilde{\mathbf{q}}} k^{-3} n_{\tilde{\mathbf{q}}/k}$. Physically, $n_{\tilde{\mathbf{q}}}$ is a genus-zero BPS index obtained from M2-branes wrapping holomorphic curves in class $\tilde{\mathbf{q}}$ in M-theory on $\tilde{X}$; mathematically, it is the integral reorganisation of rational GW invariants predicted by Gopakumar–Vafa theory. In computations, integrality of the extracted $n_{\tilde{\mathbf{q}}}$ is therefore a non-trivial consistency check, not a tautology of the GW definition.

Example 2.3 (Quintic threefold). For the quintic threefold the first few genus-zero GV invariants are

$$n_1 = 2875, \quad n_2 = 609\,250, \quad n_3 = 317\,206\,375, \quad \dots \quad (2.54)$$

The number $n_1 = 2875$ agrees with the classical count of lines on a generic quintic. Higher-degree invariants must be read as virtual, deformation-invariant counts after subtracting multiple-cover contributions, not as naive counts of smooth embedded rational curves [30]. For instance,

$$N_2 = n_2 + \frac{n_1}{2^3} = 609\,250 + \frac{2875}{8} = 609\,609.375. \quad (2.55)$$

The rational GW coefficient has been repackaged as an integer GV count plus the multiple-cover contribution of degree-one curves.

The corresponding projective period vector is obtained from the homogeneous prepotential $F(X^0, X^a) = (X^0)^2 F_0(X^a/X^0)$. In the gauge $X^0 = 1$ and $X^a = t^a$, let $F_a = \partial F_0 / \partial t^a$. Euler homogeneity gives $F_0^{\text{hom}} := \partial F / \partial X^0 = 2F_0 - t^a F_a$ in this gauge, so the A -model period vector can be written as

$$\Pi_A(t) = \begin{pmatrix} 2F_0(t) - t^a F_a(t) \\ F_a(t) \\ 1 \\ t^a \end{pmatrix} \quad (2.56)$$

up to the chosen symplectic-basis convention. The Frobenius calculation below produces the B -model representative of this same vector after applying the mirror map. Thus the comparison to keep in mind is

$$\Pi_B(\psi) \stackrel{\text{mirror}}{=} \Pi_A(t(\psi)). \quad (2.57)$$

Mirror computation for toric hypersurfaces

The direct A -model definition of the integers on $\tilde{X}$ is not the computational route used here. Instead one computes the Type IIB B -model periods on X . This is the classical side of the mirror pair: the relevant vector-multiplet geometry is encoded by periods of the holomorphic three-form Ω on X . Mirror symmetry says that, after choosing the correct integral symplectic basis and normalisation, these periods reproduce the Type IIA A -model prepotential of $\tilde{X}$.

For Batyrev hypersurfaces, the mirror pair is constructed from a dual pair of four-dimensional reflexive polytopes (Δ°, Δ) . A fine, regular, star triangulation of Δ° defines a toric fourfold V ; in the usual logic [5] the generic anti-canonical hypersurface $X \subset V$ is a smooth Calabi–Yau threefold for the Type IIB compactification. Exchanging Δ° and Δ gives the mirror $\tilde{X} \subset \tilde{V}$, whose Type IIA large volume prepotential contains the GV invariants we want. The period integral below is taken on X , but the coefficient formula is naturally indexed by ambient curve-class data of the mirror toric variety $\tilde{V}$.

The lattice relations among the monomials in the defining polynomial of X become the charge data entering the LCS coordinates. We use capital indices I, J for toric monomials and ambient toric divisors, and lowercase indices a, b, c for complex structure coordinates on X , equivalently for mirror curve/divisor bases on $\tilde{X}$. Write

$$f = \Psi^0 S_0 - \sum_I \Psi^I S_I \quad (2.58)$$

for the defining polynomial of $X \subset V$. The S_0, S_I are the monomial sections of the anticanonical bundle $-K_V$ on the original-side toric fourfold V ; they are indexed by the lattice points of the dual polytope Δ (the same polytope whose fan defines the mirror $\tilde{V}$), with S_0 the interior-point monomial corresponding to the origin. The $\Psi^0, \Psi^I \in \mathbb{C}$ are their complex coefficients; only the ratios Ψ^I / Ψ^0 enter the complex structure moduli data. The integer charge matrix Q_I^a ($a = 1, \dots, h^{1,1}(\tilde{V})$ in the ambient toric basis, restricted afterwards to the hypersurface directions used above) encodes the linear relations $\sum_I Q_I^a \nu_I = 0$ among the lattice-point vectors $\nu_I \in \Delta$; equivalently, it is the matrix of Mori-cone generators on the ambient mirror toric fourfold $\tilde{V}$, and its rows are the relations among the monomials S_I on V . Multiplicative relations among the monomials then give local gauge-invariant complex structure coordinates

$$\psi^a = \prod_I \left(\frac{\Psi^I}{\Psi^0} \right)^{Q_I^a}. \quad (2.59)$$

Interior-facet points are treated separately: the corresponding ambient divisors do not meet a generic hypersurface, and part of the toric automorphism group can be used to remove redundant

polynomial coefficients. This is one reason why the toric input must be handled as a quotient construction rather than as a list of monomials with independent coefficients.

Thus the Batyrev input plays three simultaneous roles in this mirror-period computation. It gives local complex structure coordinates on X ; it gives divisor, intersection, and Chern-class data on the mirror $\tilde{X}$; and it gives the charge matrix Q_I^a that determines the GKZ/Frobenius period computation. This is the reason the method is useful for these lectures: the same finite lattice data feed both the Type IIB period calculation and the Type IIA curve-counting calculation.

The basic period is obtained by restricting to the dense algebraic torus $(\mathbb{C}^*)^4 \subset V$ in two steps. First, the Poincaré residue defines the holomorphic three-form on X :

$$\Omega = \oint_{f=0} \left[dy^1 \wedge \cdots \wedge dy^4 \frac{1}{(2\pi i)^4 f(y)} \right]. \quad (2.60)$$

Second, the fundamental period $\varpi_0(\psi)$ is the integral of Ω over the standard small-torus cycle on $(\mathbb{C}^*)^4 \subset V$ at $\psi = 0$; equivalently, it is the constant-term Fourier mode of the right-hand side above. The resulting fundamental period, expressed as a sum over integer effective ambient curve classes in the chosen toric charge basis, is

$$\varpi_0(\psi) = \Psi^0 \int_{T^3} \Omega = \sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{V}) \cap H_2(\tilde{V}, \mathbb{Z})} c_{\tilde{\mathbf{q}}} \psi^{\tilde{\mathbf{q}}}, \quad (2.61)$$

where $T^3 \subset X$ defines a certain solution branch for $f = 0$ (the SYZ fiber [33]) and where

$$c_{\tilde{\mathbf{q}}} := \frac{\Gamma(1 + \tilde{q}_a Q_0^a)}{\prod_I \Gamma(1 + \tilde{q}_a Q_I^a)}, \quad \psi^{\tilde{\mathbf{q}}} := \prod_a (\psi^a)^{\tilde{q}_a}. \quad (2.62)$$

Here Q_I^a ($I \neq 0$) is the integer charge matrix introduced in (2.58), and Q_0^a is the charge attached to the interior monomial S_0 fixed by the Calabi–Yau condition on the toric data. The coefficient $c_{\tilde{\mathbf{q}}}$ is the basic Γ -function ratio that all the other period-series coefficients below are built from. Its analytic continuation in $\tilde{\mathbf{q}}$, obtained by replacing the integer vector $\tilde{\mathbf{q}}$ with $\tilde{\mathbf{q}} + \vec{\rho}$ for a formal auxiliary vector $\vec{\rho} = (\rho_a)$, is what allows the Frobenius derivatives below to be taken.

Initially this expression arises as a constant-term computation over movable ambient curve classes. It can be extended to the ambient Mori cone $\mathcal{M}(\tilde{V})$: outside the movable cone, denominator poles in the gamma expression kill the forbidden terms [23]. The important point is that the period is computed as a concrete toric hypergeometric series. The notation deliberately distinguishes the ambient cone $\mathcal{M}(\tilde{V})$ behind (2.61) from the hypersurface cone $\mathcal{M}(\tilde{X})$ in (2.52). The former is a computational lattice for the hypergeometric period; the latter is the geometric cone in which the enumerative invariants live. The mirror-period method of [23] works because these two pieces of information can be compared in a common toric charge basis, with the caveats about $\mathcal{M}_V^\Omega$ stated below.

The full Picard–Fuchs system is the differential system satisfied by all periods. In the toric setting it is obtained from the GKZ system determined by the same lattice relations Q_I^a . Factoring the GKZ system down to the true Picard–Fuchs system is usually expensive. The HKTY method avoids this as follows [21]. At a large complex structure point, the monodromy is maximally unipotent and is mirror to the large volume B -field monodromy of Type IIA on $\tilde{X}$. Hence the logarithmic Frobenius solutions, obtained by differentiating the term $c_{\tilde{\mathbf{q}}+\vec{\rho}} \psi^{\tilde{\mathbf{q}}+\vec{\rho}}$ of (2.61) with

respect to the auxiliary variables ρ_a before setting $\vec{\rho} = 0$, give the needed local period basis:

$$\varpi^a(\psi) = \sum_{\tilde{\mathbf{q}}} \frac{\partial_{\rho_a}}{2\pi i} \left(c_{\tilde{\mathbf{q}}+\vec{\rho}} \psi^{\tilde{\mathbf{q}}+\vec{\rho}} \right) \Big|_{\vec{\rho}=0}, \quad (2.63)$$

$$\varpi^{ab}(\psi) = \sum_{\tilde{\mathbf{q}}} \frac{\partial_{\rho_a} \partial_{\rho_b}}{(2\pi i)^2} \left(c_{\tilde{\mathbf{q}}+\vec{\rho}} \psi^{\tilde{\mathbf{q}}+\vec{\rho}} \right) \Big|_{\vec{\rho}=0}, \quad (2.64)$$

$$\varpi^{abc}(\psi) = \sum_{\tilde{\mathbf{q}}} \frac{\partial_{\rho_a} \partial_{\rho_b} \partial_{\rho_c}}{(2\pi i)^3} \left(c_{\tilde{\mathbf{q}}+\vec{\rho}} \psi^{\tilde{\mathbf{q}}+\vec{\rho}} \right) \Big|_{\vec{\rho}=0}. \quad (2.65)$$

It is useful to record once and for all the coefficient data implicit in these Frobenius solutions. Define

$$c_{\tilde{\mathbf{q}}}^a := \partial_{\rho_a} c_{\tilde{\mathbf{q}}+\vec{\rho}} \Big|_{\vec{\rho}=0}, \quad c_{\tilde{\mathbf{q}}}^{ab} := \partial_{\rho_a} \partial_{\rho_b} c_{\tilde{\mathbf{q}}+\vec{\rho}} \Big|_{\vec{\rho}=0}, \quad (2.66)$$

and assemble the formal power series

$$c^a(\psi) := \sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{V})} c_{\tilde{\mathbf{q}}}^a \psi^{\tilde{\mathbf{q}}}, \quad c^{ab}(\psi) := \sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{V})} c_{\tilde{\mathbf{q}}}^{ab} \psi^{\tilde{\mathbf{q}}}. \quad (2.67)$$

These are the gamma-function-derived power series referenced repeatedly in the period-matching formulae below. Algorithmically, $c_{\tilde{\mathbf{q}}}$, $c_{\tilde{\mathbf{q}}}^a$ and $c_{\tilde{\mathbf{q}}}^{ab}$ are independent and can be computed in parallel; cf. [23] §5.

Matching the large complex structure monodromies to the large volume monodromies fixes the constant change of basis to an integral symplectic period vector. The comparison is with the A -model vector (2.56): the first two blocks correspond to $2F_0 - \mathfrak{t}^a F_a$ and F_a , while the last two blocks correspond to 1 and the flat coordinates $\mathfrak{t}^a$. The classical data $\tilde{\kappa}_{abc}$, $\tilde{c}_a$, and $\chi(\tilde{X})$ fix the polynomial part; the remaining power series is the instanton contribution.

In the period-basis convention adopted in [23], the B -model representative of (2.56) is

$$\Pi(\psi) = \frac{1}{\varpi_0} \begin{pmatrix} \frac{1}{3!} \tilde{\kappa}_{abc} \varpi^{abc} + \frac{1}{12} \tilde{c}_a \varpi^a \\ -\frac{1}{2} \tilde{\kappa}_{abc} \varpi^{bc} + \tilde{a}_{ab} \varpi^b \\ \varpi_0 \\ \varpi^a \end{pmatrix}. \quad (2.68)$$

The ordering of components is a convention, but the lesson is coordinate-free: the polynomial part of the period vector is fixed by classical topology, and the remaining power series is where the instanton/GV data appear.

Extracting GV invariants from the periods

The flat coordinate is

$$\mathfrak{t}^a = \frac{\varpi^a}{\varpi_0} = \frac{\log \psi^a}{2\pi i} + \frac{1}{2\pi i} \frac{c^a(\psi)}{\varpi_0(\psi)}, \quad (2.69)$$

which is the *mirror map*: it identifies the B -model algebraic complex structure coordinates ψ^a on X with the A -model complexified Kähler coordinates $\mathfrak{t}^a$ on $\tilde{X}$ originally defined via $\mathfrak{t}^a = \int_{\tilde{\gamma}^a} (\tilde{B} + i\tilde{J})$ in (2.46) above. Thus

$$q^{\tilde{\mathbf{q}}} = e^{2\pi i \tilde{\mathbf{q}} \cdot \mathfrak{t}} = \psi^{\tilde{\mathbf{q}}} \exp \left(\tilde{q}_a \frac{c^a(\psi)}{\varpi_0(\psi)} \right). \quad (2.70)$$

Differentiating $F_{0,\text{inst}}$ in (2.52) and comparing with the period component $F_a = \partial F_0 / \partial \mathfrak{t}^a$ gives the master identity

$$\sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{X})} \tilde{q}_a n_{\tilde{\mathbf{q}}} \text{Li}_2 \left(\psi^{\tilde{\mathbf{q}}} \exp \left(\tilde{q}_b \frac{c^b(\psi)}{\varpi_0(\psi)} \right) \right) = \frac{1}{2} \tilde{\kappa}_{abc} \frac{\tilde{c}^{bc}(\psi) - c^b(\psi) c^c(\psi)}{\varpi_0(\psi)}. \quad (2.71)$$

Here $c^a(\psi)$ is the series defined in (2.67). The hatted series $\widehat{c}^{ab}(\psi)$ is the same Frobenius coefficient $c^{ab}(\psi)$ from (2.67), shifted by an explicit multiple of the fundamental period:

$$\widehat{c}^{ab}(\psi) := c^{ab}(\psi) - \frac{\pi^2}{6} \left(Q_0^a Q_0^b - \sum_I Q_I^a Q_I^b \right) \varpi_0(\psi). \quad (2.72)$$

Equation (2.71) is the practical bridge from periods to integers: expand both sides in powers of ψ^a , solve degree by degree, and then convert the rational Gromov–Witten coefficients into integral GV invariants using the multiple-cover formula.

There is one important cone issue. The period series is naturally written using the ambient Mori cone, but the GV invariants of interest live in the Mori cone of the hypersurface. Moreover, different triangulations of the mirror polytope Δ give different ambient Mori cones of $\widetilde{V}$, while many ambient birational transformations do not change the smooth mirror hypersurface. The computation therefore uses $\mathcal{M}_V^\cap$: an intersection of ambient Mori cones over phases for which the generic mirror hypersurface remains smooth. It is an outer approximation to $\mathcal{M}(\widetilde{X})$, not a theorem that every point in it is an effective curve class on $\widetilde{X}$. This distinction matters when interpreting zeros and truncations.

The finite truncation must also be closed under decompositions of effective classes. If a coefficient in (2.71) at class $\widetilde{\mathbf{q}}$ can receive contributions from $\widetilde{\mathbf{q}} = \widetilde{\mathbf{q}}' + \widetilde{\mathbf{q}}''$, then the coefficients at $\widetilde{\mathbf{q}}'$ and $\widetilde{\mathbf{q}}''$ must already be in the truncated computation. In the notation of [23], one may use a causal diamond

$$\diamond_{\widetilde{V}}(\widetilde{\mathbf{q}}) = \mathcal{M}_V^\cap \cap (\widetilde{\mathbf{q}} - \mathcal{M}_V^\cap) \quad (2.73)$$

and require $\diamond_{\widetilde{V}}(\widetilde{\mathbf{q}})$ to be included for every class $\widetilde{\mathbf{q}}$ whose invariant is extracted. A common practical alternative is the degree method: choose an integral grading vector $\mathbf{d}$ in the strict interior of the dual Kähler cone and retain all classes with $\mathbf{d} \cdot \widetilde{\mathbf{q}} \leq \ell$. Since degree is additive on the semigroup, this truncation is automatically closed under decompositions. Its cost scales like $\ell^{h^{1,1}}$, so at large $h^{1,1}$ one often computes modest-degree bulk data together with much deeper data on selected rays or low-dimensional faces.

The implementation of [23] stores truncated power series sparsely, orders monomials by degree, and uses hash maps to make polynomial multiplication feasible in many variables. The coefficients $c_{\widetilde{\mathbf{q}}}$, $c_{\widetilde{\mathbf{q}}}^a$, and $c_{\widetilde{\mathbf{q}}}^{ab}$ are independent and can be computed in parallel. The invariants are then extracted degree by degree: after the invariants at a fixed degree are read off, their $\text{Li}_2(q^{\widetilde{\mathbf{q}}})$ contributions are subtracted before moving to the next degree. This avoids constructing or inverting the full Picard–Fuchs system and is the reason the method scales beyond the older one- or few-parameter computations.

Aside: Toric curves and analytic checks

We introduce the phrase “toric curve” only at this point, and it needs a qualification. A generic anticanonical hypersurface $\widetilde{X} \subset \widetilde{V}_{\widetilde{\Sigma}}$ is not itself a toric variety. Let $\widehat{\widetilde{D}}_I$ denote a toric divisor of the ambient toric fourfold $\widetilde{V}_{\widetilde{\Sigma}}$, and let $\widetilde{D}_I = \widehat{\widetilde{D}}_I \cap \widetilde{X}$ denote its restriction to the hypersurface when this restriction is non-empty. Following the analytic checks of [20, 23], a toric curve class here means a curve class on $\widetilde{X}$ that is visible from the ambient toric presentation, typically a complete intersection

$$\widetilde{C}_{IJ} = \widetilde{D}_I \cap \widetilde{D}_J = \widetilde{X} \cap \widehat{\widetilde{D}}_I \cap \widehat{\widetilde{D}}_J. \quad (2.74)$$

Such curves are especially useful because some of their GV invariants can be computed directly, without mirror symmetry, and can therefore serve as checks of the period computation.

Let $\widetilde{p}_I, \widetilde{p}_J \in \Delta$ be the lattice points corresponding to $\widehat{\widetilde{D}}_I, \widehat{\widetilde{D}}_J$. A “one-face divisor” or “two-face divisor” means that the corresponding lattice point lies in the relative interior of a one- or two-dimensional face of Δ , respectively. The complete intersection $\widetilde{C}_{IJ}$ is non-empty in the simple

toric-curve analysis used here only when the segment joining $\tilde{p}_I$ and $\tilde{p}_J$ is an edge of the chosen triangulation and this edge lies in a common two-face Θ_2 of Δ . Let g be the number of interior lattice points of the polar-dual one-face $\Theta_2^\circ \subset \Delta^\circ$ (by polar duality on the reflexive pair (Δ, Δ°) : a 2-face $\Theta_2 \subset \Delta$ is dual to a 1-face $\Theta_2^\circ \subset \Delta^\circ$ with $\dim \Theta_2 + \dim \Theta_2^\circ = 3$), equivalently the genus attached to Θ_2 . On the open patch associated with Θ_2 , the mirror hypersurface is a branched cover of $g + 1$ copies of a non-compact toric Calabi–Yau threefold V_{Θ_2} . Consequently

$$\tilde{C}_{IJ} = \prod_{i=1}^{g+1} \tilde{C}_{IJ}^{(i)}, \quad \tilde{C}_{IJ}^{(i)} \simeq \mathbb{P}^1. \quad (2.75)$$

If either $\tilde{D}_I$ or $\tilde{D}_J$ is a two-face divisor, the components $\tilde{C}_{IJ}^{(i)}$ lie in distinct numerical classes. Otherwise they represent the same numerical class with multiplicity $g + 1$ [20, 23].

Under the local hypotheses used in [20, 23] for one component, the normal bundle in $\tilde{X}$ has the form

$$\mathcal{N}_{\tilde{C}_{IJ}^{(i)}/\tilde{X}} \simeq \mathcal{O}_{\mathbb{P}^1}(m) \oplus \mathcal{O}_{\mathbb{P}^1}(n), \quad m = \frac{\tilde{\kappa}_{III}}{g+1}, \quad n = \frac{\tilde{\kappa}_{IJJ}}{g+1}, \quad m+n = -2. \quad (2.76)$$

Assume $m \geq n$. For the smooth moduli spaces considered in [23], the genus-zero GV invariant is computed by

$$\text{GV}^0 = (-1)^{\dim \mathcal{M}} \chi(\mathcal{M}). \quad (2.77)$$

If the singular locus produced by shrinking the curve is compact, then the relevant moduli space is $\mathbb{P}^{m+1}$, giving

$$(-1)^{m+1}(m+2). \quad (2.78)$$

Putting the component structure together gives, for the cases covered by the compact-singular-locus assumption,

$$\text{GV}_{[\tilde{C}_{IJ}^{(i)}]}^0 = \begin{cases} (-1)^{m+1}(m+2), & \text{if either } \tilde{D}_I \text{ or } \tilde{D}_J \text{ is a two-face divisor,} \\ g+1, & \text{otherwise.} \end{cases} \quad (2.79)$$

There is one remaining case, $m = 0$ with $\tilde{D}_J$ a one-face divisor. Then $\tilde{D}_J$ is a $\mathbb{P}^1$ -fibration over a curve of genus g' , where g' equals the number of interior lattice points of the dual two-face of Δ° containing the relevant edge, and the moduli space of the fibre class is that base curve. The corresponding invariant is $2g' - 2$.

These formulas are not a substitute for the full mirror computation: they apply to a special class of ambient-visible rational curves. Their value for the lectures is conceptual and practical. They show what kind of curve geometry the integers detect, and they provide local checks on the GV dictionaries used in lecture 3 for flop navigation and wall crossing in the (torically) extended Kähler cone. These toric curve classes include the Mori-extremal classes whose volumes can shrink at chamber walls of the secondary fan, so their GV invariants are the enumerative data that survive into the flop transitions of Lecture 3 and into the explicit constructions it feeds.

2.6 Application: perturbatively flat vacua from periods and GV invariants

This final section is the promised application of the period and GV machinery developed above. Up to this point, the lecture has explained how toric mirror symmetry computes the period vector and how the GV integers enter its instanton expansion. We now insert that computed period vector into the flux equations of motion. The aim is not yet a full vacuum-energy construction; it is to show, in one explicit class of examples, how enumerative data become numerical input for a flux superpotential.

2.6.1 From periods to equations of motion: perturbatively flat vacua

We have now reached the first payoff of the period calculation. Near an LCS point, mirror symmetry gives an explicit series expansion for the period vector $\Pi(z)$, including the worldsheet-instanton corrections controlled by GV invariants. Once the flux integers are chosen, the flux superpotential is therefore a computable holomorphic function. Schematically, if f and h collect the integral F_3 - and H_3 -flux quanta, then

$$W_{\text{flux}}(\tau, z) = (f - \tau h) \cdot \Pi(z), \quad (2.80)$$

where the dot denotes the symplectic pairing in the chosen integral period basis. The next problem is no longer to compute periods. It is to solve the equations of motion for the axio-dilaton and complex structure moduli.

For constant scalar fields, the general equations of motion are the critical-point equations for the scalar potential,

$$\partial_{\Phi^I} V = 0, \quad (2.81)$$

with $\Phi^I = (\tau, z^a, T_i)$. In the supersymmetric flux sector for (τ, z^a) , it is enough to solve the F -term equations

$$D_\tau W_{\text{flux}} = 0, \quad D_a W_{\text{flux}} = 0, \quad (2.82)$$

with $D_I W = \partial_I W + (\partial_I K)W$. These conditions are equivalent to $\star G_3 = iG_3$ with $\star$ corresponding to the Hodge-star on X and $G_3 = F_3 - \tau H_3$. For $I = a$, the relevant K is precisely the Weil–Petersson Kähler potential (2.21), and the metric entering the scalar potential is (2.20). Thus the period computation supplies both the function W_{flux} and the geometry with respect to which its covariant derivatives are taken. The equations (2.82) are the four-dimensional equations of motion restricted to the supersymmetric critical-point ansatz for (τ, z^a) . For a generic flux vector, (2.82) is a coupled system of transcendental equations, because the periods contain logarithms and instanton series. They are Diophantine because the unknown flux quanta are required to be integers. With an integral symplectic basis fixed, the fluxes are integer vectors, so the equations (2.82) impose algebraic constraints on integer coefficients rather than on arbitrary real parameters. In the LCS regime, the leading polynomial part makes this especially explicit: one is asking for integer flux vectors satisfying a system of linear or polynomial relations, often together with a tadpole bound. This is difficult because integer solutions are sparse and discontinuous; unlike ordinary numerical minimisation, one cannot move infinitesimally in flux space, and the search grows rapidly with $b_3(X)$ and the allowed tadpole range. In practice one therefore usually solves the system (2.82) numerically.

Instead, we make use of the *perturbatively flat vacuum mechanism* of Demirtas, Kim, McAllister and Moritz [28]. Such perturbatively flat vacua are special because part of the problem becomes *algebraic*. The idea is to choose the fluxes so that the polynomial LCS approximation to W_{flux} has a flat solution locus on which both W_{flux} and its first derivatives vanish. The exponentially small instanton terms, whose coefficients are the GV invariants computed above, then lift this flat direction. This produces a computable one-variable problem and can leave an exponentially small value for

$$W_0 = \langle W_{\text{flux}} \rangle. \quad (2.83)$$

Thus the construction should be read as a controlled shortcut through the flux equations of motion: most flux choices lead to a numerical root-finding problem, while the PFV flux choices first solve the polynomial LCS equations exactly and then use the instanton-corrected periods to stabilise the remaining direction. This is why PFVs are useful for later: small $|W_0|$ pushes the (SUSY) Kähler modulus minimum to large divisor volume. But the logical order in this lecture is simply

$$\text{periods at LCS} \implies \text{flux equations of motion} \implies \text{PFVs as analytic foothold.} \quad (2.84)$$

Aside: The tadpole constraint on flux choices*

The Diophantine problem is also bounded. In a compact orientifold, three-form flux carries D3-brane charge and must fit inside the global D3-charge budget. With the localised O-plane and D7-brane curvature contributions moved to the right-hand side, the standard convention used in the GKP/F-theory formulation is:

Proposition 2.4 (D3 tadpole condition). *For a Type IIB O3/O7 compactification with F-theory lift Y_4 (an elliptically fibred Calabi–Yau fourfold), the fluxes and net mobile D3-charge obey*

$$N_{\text{D3}} - N_{\overline{\text{D3}}} + \frac{1}{2} \int_X H_3 \wedge F_3 = \frac{\chi(Y_4)}{24}. \quad (2.85)$$

Equivalently, the flux contribution $N_{\text{flux}} = \int_X H_3 \wedge F_3$ is restricted by the finite D3-charge budget after a choice of localised sources. Signs vary with the ordering convention for F_3 and H_3 ; the convention above matches the use of $G_3 = F_3 - \tau H_3$ in these notes. See [46, 42].

The PFV computation in LCS coordinates

We now spell out the calculation, because this is the first point in the lectures where the period technology from §2.5 directly produces a vacuum ingredient. Near an LCS point, choose flat complex structure coordinates z^a , $a = 1, \dots, h^{2,1}(X)$. These are the B -model counterparts of the mirror complexified Kähler coordinates $\mathfrak{t}^a$ used in the mirror-period discussion of §2.5. In the symplectic period basis adapted to the LCS prepotential (the basis underlying the period vector Π of (2.68), with slot ordering $(\mathcal{F}_0, \mathcal{F}_a, X^0, X^a)^\top$), write the flux vectors as

$$\vec{f} = (R_0, R_a, 0, M^a)^\top, \quad \vec{h} = (0, K_a, 0, \mathbf{0})^\top, \quad R_0, R_a, M^a, K_a \in \mathbb{Z}. \quad (2.86)$$

The bold $\mathbf{0}$ in the magnetic slot of $\vec{h}$ is the vector zero (matching the vector M^a in the same slot of $\vec{f}$), not a scalar zero. Let $\tilde{\kappa}_{abc}$, $\tilde{a}_{ab}$, and $\tilde{c}_a$ denote the cubic, quadratic, and linear polynomial data of the LCS prepotential in this basis. Up to the common overall normalisation convention for W_{flux} , the period expansion gives

$$W_{\text{flux}} = W_{\text{poly}} + W_{\text{exp}}, \quad (2.87)$$

with

$$W_{\text{poly}}(z, \tau) = \frac{1}{2} M^a \tilde{\kappa}_{abc} z^b z^c - \tau K_a z^a + (R_a - \tilde{a}_{ab} M^b) z^a + \left(R_0 - \frac{1}{24} M^a \tilde{c}_a \right), \quad (2.88)$$

$$W_{\text{exp}}(z) = -\frac{1}{(2\pi)^2} \sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{X})} n_{\tilde{\mathbf{q}}} \tilde{q}_a M^a \text{Li}_2\left(e^{2\pi i \tilde{q}_a z^a}\right). \quad (2.89)$$

Here $\tilde{X}$ is the mirror used to compute the periods, and $n_{\tilde{\mathbf{q}}}$ are the relevant genus-zero GV invariants of curve classes on $\tilde{X}$. Thus the same integers that entered the instanton part of the prepotential in (2.52) now appear as coefficients in the flux superpotential.

The point of the special flux ansatz is that the polynomial part can be made to vanish before using the exponentially small terms. If

$$R_a = \tilde{a}_{ab} M^b \in \mathbb{Z}, \quad R_0 = \frac{1}{24} M^a \tilde{c}_a \in \mathbb{Z}, \quad (2.90)$$

then the constant and linear terms in W_{poly} are absent. Since $\tilde{a}_{ab}$ is in general half-integral (see (2.50)), the integrality conditions are best viewed as congruence conditions on M . Projecting to the M -variables, the allowed choices are the kernel of the homomorphism

$$\mathbb{Z}^{h^{2,1}(X)} \longrightarrow (\mathbb{Q}/\mathbb{Z})^{h^{2,1}(X)+1}, \quad M \longmapsto \left(\tilde{a}_{ab} M^b, \frac{1}{24} \tilde{c}_a M^a \right) \bmod \mathbb{Z}. \quad (2.91)$$

This kernel always contains 0 and is a finite-index sublattice of the integer flux lattice (the codomain is a finite abelian group, since $\tilde{a}_{ab}$ and $\tilde{c}_a/24$ are rational with bounded denominators). For any M^a in this kernel, R_a, R_0 are then fixed by (2.90). Define

$$N_{ab} := \tilde{\kappa}_{abc} M^c, \quad p^a := N^{ab} K_b, \quad (2.92)$$

where N^{ab} is the inverse of N_{ab} . Via mirror symmetry the rational vector $p = p^a \tilde{\omega}_a$ is viewed as a direction in the mirror Kähler cone. The ordinary derivatives of the polynomial superpotential are

$$\partial_{z^a} W_{\text{poly}} = N_{ab} z^b - \tau K_a, \quad \partial_\tau W_{\text{poly}} = -K_a z^a. \quad (2.93)$$

Thus $\partial_{z^a} W_{\text{poly}} = 0$ forces the complex line

$$z^a = p^a \tau. \quad (2.94)$$

On this line one has

$$W_{\text{poly}} = -\frac{1}{2} \tau^2 K_a p^a, \quad \partial_\tau W_{\text{poly}} = -\tau K_a p^a. \quad (2.95)$$

Consequently the polynomial contribution satisfies

$$W_{\text{poly}} = \partial_\tau W_{\text{poly}} = \partial_{z^a} W_{\text{poly}} = 0 \quad (2.96)$$

along $z^a = p^a \tau$, provided that

$$K_a p^a = K_a N^{ab} K_b = 0. \quad (2.97)$$

Since $W_{\text{poly}} = 0$ on this locus, the covariant F -terms $D_I W_{\text{poly}}$ reduce to the ordinary derivatives displayed above.

The PFV conditions are therefore the following arithmetic and geometric requirements:

$$\det N \neq 0, \quad p = p^a \tilde{\omega}_a \in \mathcal{K}(\tilde{X}), \quad K_a p^a = 0, \quad \tilde{a}_{ab} M^b \in \mathbb{Z}, \quad M^a \tilde{c}_a \in 24\mathbb{Z}. \quad (2.98)$$

The condition $p = p^a \tilde{\omega}_a \in \mathcal{K}(\tilde{X})$ is not cosmetic: it says that the line $z^a = p^a \tau$ remains in the LCS region when $\text{Im } \tau$ is large.

Diophantine viewpoint. For a mathematician, (2.98) is perhaps the most interesting line in the construction. The unknowns are not continuous parameters but integral flux vectors M^a, K_a , together with the induced rational vector $p^a = N^{ab} K_b$. The equations combine a quadratic null condition $K_a N^{ab} K_b = 0$, congruence conditions coming from the integral symplectic basis, the open cone condition $p \in \mathcal{K}(\tilde{X})$, and the D3-tadpole bound. Thus the small- W_0 problem is, at this stage, a constrained Diophantine search whose coefficients are the classical and enumerative data of the Calabi–Yau. The physics then asks for solutions that additionally place the instanton expansion in a controlled region.

After imposing (2.98), the only surviving contribution along the flat direction is

$$W_{\text{eff}}(\tau) = -\frac{1}{(2\pi)^2} \sum_{\tilde{\mathbf{q}} \in \mathcal{M}(\tilde{X})} n_{\tilde{\mathbf{q}}} \tilde{q}_a M^a \text{Li}_2\left(e^{2\pi i \tilde{q}_a p^a \tau}\right). \quad (2.99)$$

This is the promised bridge from the mirror-period computation to flux vacua: once the truncated GV dictionary is known, (2.99) is an explicit computable holomorphic function of one variable. The racetrack discussion below is the elementary mechanism by which several leading terms in this series can stabilise τ and leave an exponentially small value of W_0 .

The racetrack mechanism, made concrete

The PFV mechanism produces a residual effective superpotential along the flat direction τ (the variable already used in (2.99)) of the form

$$W_{\text{eff}}(\tau) = c \left(e^{2\pi i p_1 \tau} + A e^{2\pi i p_2 \tau} \right) + \dots, \quad (2.100)$$

where the rationals $p_1 = \tilde{q}_a^{(1)} p^a$ and $p_2 = \tilde{q}_a^{(2)} p^a$ are built from the two leading curve classes $\tilde{\mathbf{q}}^{(1)}, \tilde{\mathbf{q}}^{(2)}$ and the rational vector $p^a = N^{ab} K_b$ of (2.92); they need not be integers. The prefactors c, A are determined by the leading genus-zero GV invariants of the mirror $\tilde{X}$ that enter (2.89). Solving $\partial_\tau W_{\text{eff}} = 0$ for τ and substituting back, the residual value $|W_0|$ inherits the exponential smallness of the worldsheet-instanton coefficients c, A together with a fractional power of their ratio controlled by $1/|p_2 - p_1|$. For example, if phases are aligned and $p_2 > p_1 > 0$ along the positive-imaginary saddle, the equation $\partial_\tau W_{\text{eff}} = 0$ gives

$$\text{Im } \tau \simeq \frac{1}{2\pi(p_2 - p_1)} \log \left| \frac{Ap_2}{cp_1} \right|, \quad (2.101)$$

and substituting back gives schematically

$$|W_0| \simeq |c| \left| 1 - \frac{p_1}{p_2} \right| \left| \frac{cp_1}{Ap_2} \right|^{p_1/(p_2 - p_1)}. \quad (2.102)$$

The prefactors and phases depend on the chosen two leading terms, but the combinatorial choice of flux quanta controls both the exponent and the integer coefficients entering through the GV invariants in c, A ; see [28] for the exact dependence.

Example 2.5 (Two-modulus PFV example [28]). Consider the degree-18 hypersurface in $\mathbb{P}_{[1,1,1,6,9]}^4$. The full Calabi–Yau threefold has $h^{2,1} = 272$, but one restricts to the $G = \mathbb{Z}_6 \times \mathbb{Z}_{18}$ -invariant locus of [47]. For G -invariant fluxes the F-term equations are consistently restricted to this locus, and the relevant periods are those of an effective two-modulus Greene–Plesser mirror [48]. This is the example used in [28]; see also [26] for the two-parameter mirror-symmetry data. The orientifold choice used there has D3-charge budget $Q_{\text{D3}} = 138$. We shall write the two LCS coordinates as z^1, z^2 .

In the notation of (2.49), the classical data are

$$\tilde{\kappa}_{111} = 9, \quad \tilde{\kappa}_{112} = 3, \quad \tilde{\kappa}_{122} = 1, \quad (2.103)$$

together with

$$\tilde{a} = \frac{1}{2} \begin{pmatrix} 9 & 3 \\ 3 & 0 \end{pmatrix}, \quad \tilde{c} = \begin{pmatrix} 102 \\ 36 \end{pmatrix}. \quad (2.104)$$

The first instanton corrections to the prepotential are, in the normalisation of [7] §6.2,

$$(2\pi i)^3 F_{\text{inst}} = F_1 + F_2 + \dots, \quad (2.105)$$

where

$$F_1 = -540 e^{2\pi i z^1} - 3 e^{2\pi i z^2}, \quad (2.106)$$

$$F_2 = -\frac{1215}{2} e^{4\pi i z^1} + 1080 e^{2\pi i(z^1 + z^2)} + \frac{45}{8} e^{4\pi i z^2}. \quad (2.107)$$

The important numerical feature is already visible in F_1 : the two one-instanton coefficients are 540 and 3. The racetrack will use this hierarchy together with a flux choice that makes the two exponents close.

Let $M = (M^1, M^2)^\top$ and $K = (K_1, K_2)^\top$ be the two flux vectors in the PFV ansatz (2.86). With the intersection numbers displayed above, the null condition $K_a N^{ab} K_b = 0$ becomes the explicit Diophantine relation

$$M^1 = \frac{M^2 K_2 (2K_1 - 3K_2)}{(K_1 - 3K_2)^2}. \quad (2.108)$$

When this relation is obeyed and the integrality conditions (2.90) hold, the flat PFV direction is

$$\begin{pmatrix} z^1 \\ z^2 \end{pmatrix} = \tau \frac{K_1 - 3K_2}{M^2} \begin{pmatrix} -K_2/K_1 \\ 1 \end{pmatrix}. \quad (2.109)$$

Thus the problem has become arithmetic: one searches for integer vectors M, K satisfying this rational equation, the congruence conditions needed for $R_0, R_a \in \mathbb{Z}$, the Kähler-cone condition on p , and the tadpole bound.

Substituting the flat direction into the one-instanton terms gives a two-term racetrack

$$W_{\text{eff}}(\tau) = c (e^{2\pi i p_1 \tau} + A e^{2\pi i p_2 \tau}) + \dots, \quad (2.110)$$

where $p_1 = p^1$ and $p_2 = p^2$ for the two leading curve classes. Solving $\partial_\tau W_{\text{eff}} = 0$ gives

$$e^{2\pi i (p_1 - p_2) \langle \tau \rangle} = -A \frac{p_2}{p_1}, \quad \langle \tau \rangle = \frac{1}{2\pi i} \frac{1}{p_1 - p_2} \log \left(-A \frac{p_2}{p_1} \right), \quad (2.111)$$

and therefore

$$W_{\text{eff}}(\langle \tau \rangle) = c \frac{p_2 - p_1}{p_2} \left(-A \frac{p_2}{p_1} \right)^{p_1 / (p_1 - p_2)}. \quad (2.112)$$

The small number is produced when $|p_1 - p_2| \ll p_2$: the two instantons then compete at a large value of $\text{Im } \tau$, and the value of the superpotential is exponentially suppressed.

The explicit flux choice used in [28] is

$$M = \begin{pmatrix} -16 \\ 50 \end{pmatrix}, \quad K = \begin{pmatrix} 3 \\ -4 \end{pmatrix}. \quad (2.113)$$

The associated flat-direction vector is

$$p = \begin{pmatrix} 2/5 \\ 3/10 \end{pmatrix}, \quad z^1 = \frac{2}{5} \tau, \quad z^2 = \frac{3}{10} \tau. \quad (2.114)$$

The flux contribution to the D3 tadpole is

$$Q_{\text{flux}} = -\frac{1}{2} M^a K_a = 124 \leq 138. \quad (2.115)$$

The two-instanton racetrack in (2.100) has

$$A = -\frac{5}{288}, \quad c = -\sqrt{\frac{2}{\pi}} \frac{8640}{(2\pi i)^3}. \quad (2.116)$$

Solving the full F-term equations gives

$$\langle \tau \rangle = 6.856 i, \quad \langle z^1 \rangle = 2.742 i, \quad \langle z^2 \rangle = 2.057 i, \quad (2.117)$$

and

$$|W_0| = 2.037 \times 10^{-8}. \quad (2.118)$$

The accompanying GitHub repository contains a demo notebook, `jaxvacua_demo.ipynb` in the `notebooks/` directory, which reconstructs this PFV input from the flux vectors above and solves the corresponding F-term equations numerically. The two-instanton terms displayed above are smaller by $O(10^{-5})$ at this solution, so the two-instanton racetrack already captures the scale of the answer.

What this example demonstrates. This example shows how mirror pairs constructed from polytope triangulations can turn finite combinatorial input into integer enumerative data, and how those integers can then produce very small numbers in string-theoretic quantities. The small value of W_0 is not obtained by tuning continuous parameters: it comes from GV invariants in the instanton expansion, combined with integer flux quanta through the PFV Diophantine conditions. This is highly non-trivial, and it is one of the key inputs for the next lecture, where the goal is to compute the full vacuum energy; the same suppression mechanism is what makes parametrically tiny scales available in the effective theory.

Lecture 2 punchlines. The second lecture has one computational message. To use fluxes in a Calabi–Yau compactification one must compute the periods of the holomorphic three-form, because

$$W_{\text{flux}} = \int_X \Omega \wedge G_3 \quad (2.119)$$

is a flux-weighted period integral. The integers specifying F_3 and H_3 choose the linear combination; mirror symmetry and toric geometry compute the functions that appear in that linear combination.

The mirror computation translates the same period vector into the large volume A -model prepotential on $\tilde{X}$. Its instanton part is organised by genus-zero GV invariants, so these integers enter the four-dimensional superpotential as actual coefficients, not merely as auxiliary enumerative data. The pipeline established in the lecture is therefore

$$\begin{aligned} \text{toric hypersurface data} &\implies \text{periods and mirror map} \\ &\implies \text{GV invariants} \implies W_{\text{flux}}. \end{aligned} \quad (2.120)$$

The PFV computation at the end of the lecture is the first application of this pipeline. It chooses fluxes so that the polynomial LCS contribution to W_{flux} has a flat solution locus, and then uses the GV-controlled instanton terms to lift that locus and produce a small value of W_0 . This is precisely the calculation illustrated in the companion notebook `notebooks/jaxvacua_demo.ipynb`. Lecture 3 starts from this flux-sector output and asks what is still required to obtain an actual vacuum energy.

A General conventions and notation

We collect the conventions used throughout. This is a reference section for the written notes, not a preliminary lecture. Each main lecture repeats the minimum notation and equations it needs. The conventions follow [7], § 2.2, and have been streamlined for the present audience.

A.1 Indices and spacetime

The ten dimensions of the ambient string-theoretic spacetime are indexed by capital Latin letters $M, N, \dots = 0, \dots, 9$. The ansatz of physical interest splits these into four “external” dimensions $\mu, \nu, \dots = 0, 1, 2, 3$ corresponding to the four-dimensional spacetime that our cosmology lives in, and six “internal” dimensions $m, n, \dots = 4, \dots, 9$ corresponding to a compact real 6-manifold X :

$$ds_{10}^2 = e^{2A(y)} g_{\mu\nu}^{(4)}(x) dx^\mu dx^\nu + e^{-2A(y)} g_{mn}^{(6)}(y) dy^m dy^n. \quad (\text{A.1})$$

The factor $e^{2A(y)}$ is the *warp factor*; we will set it to unity at first and reinstate it in Lecture 3. We use signature $(-, +, +, \dots, +)$ for $g^{(10)}$.

A.2 Differential forms and the Hodge decomposition

For a real C^∞ manifold M of dimension D we write $\Omega^r(M)$ for the space of smooth r -forms, $d : \Omega^r \rightarrow \Omega^{r+1}$ for the exterior derivative, and $\star : \Omega^r \rightarrow \Omega^{D-r}$ for the Hodge star with respect to

a Riemannian metric g . The adjoint $d^\dagger = (-1)^{Dr+D+1} \star d \star$ and the Laplacian $\Delta = dd^\dagger + d^\dagger d$ act on $\Omega^r(M)$.

We denote by $\mathcal{H}^r(M)$ the space of harmonic r -forms. For M compact orientable, Hodge’s theorem identifies

$$H^r(M, \mathbb{R}) \cong \mathcal{H}^r(M). \quad (\text{A.2})$$

We write $b_r(M) = \dim_{\mathbb{R}} H^r(M, \mathbb{R})$ for the Betti numbers.

A.3 Complex manifolds and Dolbeault cohomology

A *complex manifold* of complex dimension k is a real $2k$ -manifold with complex coordinate charts whose transition functions are holomorphic. The tangent bundle splits $TM \otimes \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$, and differential forms decompose into (p, q) -types,

$$\Omega^r(M, \mathbb{C}) = \bigoplus_{p+q=r} \Omega^{p,q}(M), \quad (\text{A.3})$$

with $d = \partial + \bar{\partial}$ on a Hermitian manifold and $\bar{\partial}^2 = 0$. The $\bar{\partial}$ -Laplacian $\Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^\dagger + \bar{\partial}^\dagger\bar{\partial}$ defines harmonic (p, q) -forms; we write $\mathcal{H}_{\bar{\partial}}^{p,q}(M)$ for the space and $h^{p,q}(M) = \dim \mathcal{H}_{\bar{\partial}}^{p,q}(M)$ for the *Hodge numbers*.

On a Kähler manifold (Hermitian metric whose Kähler form J is closed) one has the identity $\Delta_d = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}$, hence

$$b_r(M) = \sum_{p+q=r} h^{p,q}(M), \quad h^{p,q} = h^{q,p}, \quad h^{p,q} = h^{k-p, k-q}. \quad (\text{A.4})$$

A.4 Calabi–Yau threefolds

There is more than one convention for “Calabi–Yau”; we record ours explicitly. *In these notes a Calabi–Yau n -fold is a compact Kähler manifold X of complex dimension n with trivial canonical bundle $K_X \cong \mathcal{O}_X$ and $\pi_1(X) = 0$.* The convention matches that used in the candidate-dS literature [7, 1]; the simple-connectedness clause is convenient but not standard across the differential-geometry literature (Joyce, for instance, defines a CY n -fold as a compact Kähler manifold with $\text{Hol}(X) = \text{SU}(n)$, which implies $\pi_1(X)$ finite but not necessarily trivial). Under our convention, the following standard structural facts hold; in each case the implication should be read in the indicated direction.

- Triviality of K_X is equivalent to the existence of a nowhere-vanishing holomorphic $(n, 0)$ -form $\Omega \in H^0(X, K_X) = H^{n,0}(X)$.
- Yau’s theorem (Thm. 1.7 below) gives the existence and uniqueness of a Ricci-flat Kähler metric in each Kähler class.
- For this Ricci-flat metric, the restricted holonomy (which, under $\pi_1(X) = 0$, equals the full holonomy) satisfies $\text{Hol}(X) \subseteq \text{SU}(n)$; equality $\text{Hol}(X) = \text{SU}(n)$ requires further input, namely that X is irreducible (not a Riemannian product) and not hyperkähler, in which case Berger’s classification (Theorem 1.2) gives $\text{Hol}(X) = \text{SU}(n)$.

For $n = 3$ the Hodge diamond and the resulting Euler characteristic $\chi(X) = 2(h^{1,1}(X) - h^{2,1}(X))$ are stated in §1.6 (Proposition ‘CY3 Hodge diamond’); we do not repeat them here.

A.5 Toric notation (reminder; not derived)

We use the standard toric-geometry conventions of [4]. For a complete simplicial fan $\Sigma \subset N_{\mathbb{R}}$ (with $N \cong \mathbb{Z}^d$) we write V_{Σ} for the associated toric variety. Prime toric divisors D_{ρ} correspond to rays $\rho \in \Sigma(1)$. The Chow group $A^1(V_{\Sigma}) \otimes \mathbb{R}$ has a distinguished cone $\mathcal{K}(V_{\Sigma})$ (the Kähler cone) dual to the Mori cone $\mathcal{M}(V_{\Sigma}) := \overline{\text{NE}}(V_{\Sigma})$ (the closure of the cone generated by classes of effective 1-cycles; see Definition ??). The notation $\mathcal{M}$ is used in the lectures to avoid confusing the Mori cone with the nef cone; the standard algebraic-geometry notation is displayed at the point of

definition. For a 4-dimensional reflexive polytope $\Delta^\circ \subset N_{\mathbb{R}}$ (with polar dual $\Delta \subset M_{\mathbb{R}}$, $M = N^\vee$), the Batyrev construction [5] produces a Calabi–Yau threefold X as the generic anticanonical hypersurface in the ambient toric fourfold $V_{\Sigma_{\mathcal{T}}}$ associated to a fine, regular, star triangulation (FRST) $\mathcal{T}$ of Δ° (Appendix ??). The ambient $V_{\Sigma_{\mathcal{T}}}$ has at worst $\mathbb{Q}$ -factorial terminal Gorenstein singularities, located in cones of codimension ≥ 4 in $V_{\Sigma_{\mathcal{T}}}$ (i.e. at isolated points), which generically do not meet the smooth hypersurface X .

We write $\kappa_{abc} = D_a \cdot D_b \cdot D_c$ for the triple intersection numbers of (pullbacks of) prime toric divisors on X . For the Calabi–Yau hypersurface X , the Kähler cone is denoted $\mathcal{K}(X) \subset H^{1,1}(X, \mathbb{R})$; the *extended Kähler cone* $\tilde{\mathcal{K}}(X)$ is the union of $\mathcal{K}(X')$ over all birational models X' reached by chains of flops, glued along their common flop walls (see Proposition ?? and Appendix ??).

The mirror Calabi–Yau threefold of X is denoted $\tilde{X}$ (Batyrev construction, Appendix ??); we do not use the X° notation. Note that the tilde has two unrelated uses in these notes: the mirror manifold $\tilde{X}$ and the extended Kähler cone $\tilde{\mathcal{K}} = \tilde{\mathcal{K}}$.

A.6 Conventions for field-theory quantities

We work in $\hbar = c = 1$ units, and adopt $M_{\text{Pl}} = 1$, where the four-dimensional reduced Planck mass has dimensionful value $M_{\text{Pl}} \simeq 2.4 \times 10^{18}$ GeV. A *field* on a spacetime manifold is a geometric object allowed to vary from point to point: for example a function, a differential form, a connection, or a metric. An *action* is a functional of these fields,

$$S[\Phi, g] = \int_M d^{\dim M} x \sqrt{|g|} \mathcal{L}(\Phi, g, \nabla \Phi, \dots), \quad (\text{A.5})$$

where Φ denotes all non-metric fields and $\mathcal{L}$ is the Lagrangian density. Varying S with respect to Φ gives the field equations. These field equations are what physicists call the *equations of motion*: the differential equations satisfied by classical field configurations that are stationary points of the action. Varying S with respect to the metric gives the stress-energy tensor

$$T_{MN} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter+source}}}{\delta g^{MN}}. \quad (\text{A.6})$$

Thus a schematic ten-dimensional action of the form

$$S^{(10)} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g} (R^{(10)} + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{source}}) \quad (\text{A.7})$$

gives the Einstein equation

$$R_{MN}^{(10)} - \frac{1}{2}g_{MN}R^{(10)} = \kappa_{10}^2 T_{MN}. \quad (\text{A.8})$$

Taking the trace in 10 dimensions rewrites this as $R_{MN}^{(10)} = \kappa_{10}^2 (T_{MN} - \frac{1}{8}g_{MN}T)$, with $T = T^K_K$.

A four-dimensional field is called *light* relative to an effective description with cutoff Λ if its mass is well below Λ , so it is retained as a dynamical variable. Fields with masses comparable to or above Λ are integrated out and affect the light theory only through corrections to its couplings. In a compactification, harmonic zero modes are massless at leading order and hence light before fluxes, branes, or quantum corrections generate a potential.

The ten-dimensional gravitational coupling κ_{10}^2 is normalised so that the leading-order action takes the form

$$S^{(10)} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g_{10}} \left(R^{(10)} - \frac{1}{2}|\partial\tau|_\tau^2 - \frac{1}{2}|G_3|^2 - \frac{1}{4}|\tilde{F}_5|^2 + \dots \right), \quad (\text{A.9})$$

with $\tau = C_0 + i e^{-\phi}$ the axio-dilaton, $G_3 = F_3 - \tau H_3$ the complexified three-form, and $\tilde{F}_5$ the self-dual five-form. This schematic expression fixes notation only; Lecture 2 uses the four-dimensional period and superpotential data rather than the full ten-dimensional pseudo-action. The norm $|\partial\tau|_\tau^2$ uses the hyperbolic metric on the upper half-plane.

Remark A.1 (A note on physics terms). Throughout these lectures, “supergravity” (SUGRA) denotes the low-energy field-theory approximation to a chosen string background: an action-based field theory on a Lorentzian manifold of fixed dimension, retaining the massless fields and their leading interactions. “Supersymmetry” (SUSY) denotes a $\mathbb{Z}_2$ -graded extension of the Poincaré algebra by spinorial generators; an $\mathcal{N} = 1$ theory in D dimensions has one such spinor generator (so 4 real supercharges in $D = 4$). “Flux” denotes a properly quantised closed differential form, or more precisely its integral cohomology class. In the supersymmetric geometric approximation, a “brane” is represented by a calibrated submanifold of X carrying gauge / matter degrees of freedom. Precise definitions appear in Lectures 2 and 3.

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